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Small improvements to the Ball–Rivoal theorem and its p-adic variant

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Abstract

We prove that the dimension of the $\mathbb{Q}$-linear span of the Riemann zeta values $\zeta(3), \zeta(5), \ldots, \zeta(s-1)$ is at least $(1.119 \cdot \log s)/(1+\log 2)$ for any sufficiently large even integer $s \geqslant s_0$. This result slightly refines the theorem of Rivoal (2000) or Ball–Rivoal (2001). The proof incorporates the arithmetic factor Φ_n introduced by Zudilin (2001) into a slight generalization of the Ball–Rivoal hypergeometric construction. The asymptotic impact of the arithmetic factor Φ_n was not anticipated prior to this work. Although this result is subsumed by a much stronger recent development by Fischler (2026), our approach yields a new result in the p-adic setting, slightly refining a theorem of Sprang (2020). We prove that the dimension of the $\mathbb{Q}$-linear span of the p-adic zeta values $\zeta_p(3), \zeta_p(5), \ldots, \zeta_p(s-1)$ is at least $(1.119 \cdot \log s)/(1+\log 2)$ for any prime p and any sufficiently large even integer $s \geqslant s_0(p)$.

1 Introduction

The arithmetic nature of special values of the Riemann zeta function $\zeta(\cdot)$ has attracted much attention. The values of $\zeta(\cdot)$ at positive even integers are well understood thanks to Euler's formula, which shows that $\zeta(2k)$ is a non-zero rational multiple of π^{2k} for any positive integer k. In contrast, the values of $\zeta(\cdot)$ at positive odd integers greater than one (referred to as *odd zeta values* for brevity) are more mysterious. It is conjectured that π and all the odd zeta values are algebraically independent over $\mathbb{Q}$. Toward this conjecture, Rivoal in 2000 (or Ball–Rivoal in 2001) made the following breakthrough:

Theorem 1.1 (Rivoal [14], 2000; Ball–Rivoal [2], 2001)**.** *We have*

$$\dim_{\mathbb{Q}} \operatorname{Span}_{\mathbb{Q}} (1, \zeta(3), \zeta(5), \ldots, \zeta(s)) \geqslant \frac{1-o(1)}{1+\log 2} \cdot \log s$$

as the odd integer $s \to \infty$.

In 2020, Sprang established a p-adic version of the Ball–Rivoal theorem. He proved the following theorem:

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Theorem 1.2 (Sprang [16], 2020)**.** *Let p be any prime number. We have*

$$\dim_{\mathbb{Q}} \operatorname{Span}_{\mathbb{Q}} (1, \zeta_p(3), \zeta_p(5), \dots, \zeta_p(s)) \geqslant \frac{1 - o(1)}{2(1 + \log 2)} \cdot \log s$$

as the odd integer $s \to \infty$.

Here $\zeta_p(s)$ (for an odd integer $s > 1$) is the p-adic zeta value. We will recall basic facts about p-adic zeta values in §7. Note that there is an additional factor $1/2$ in Sprang's theorem compared to the Ball–Rivoal theorem. Sprang subsequently removed this additional $1/2$ factor in unpublished work.

The proof of the Ball–Rivoal theorem involves explicit rational functions defined as follows:

$$R_n(t) = n!^{s-2r} \cdot \frac{(t - rn)_{rn} \cdot (t + n + 1)_{rn}}{(t)^s_{n+1}}, \quad n = 2, 4, 6, \dots,$$

where $r = \lfloor s/\log^2 s \rfloor$ is a positive integer, and $(x)_k = x(x+1)(x+2)\cdots(x+k-1)$ denotes the Pochhammer symbol. The quantity $S_n = D_n^s \cdot \sum_{\nu=1}^{\infty} R_n(\nu)$ represents a linear form in 1 and odd zeta values with integer coefficients, where D_n is the least common multiple of $1, 2, \dots, n$.

In 2001, Zudilin [17] observed that if we consider different lengths for each Pochhammer symbol appearing in $R_n(t)$, then the coefficients of the resulting linear form have a large common divisor Φ_n. Specifically, Zudilin considered rational functions of the form:

$$\begin{aligned} R_n(t) = {} & (\text{some factor}) \cdot (2t + Mn) \\ & \times \frac{\prod_{j=1}^{b} (t - rn)_{(r+\delta_{-j})n} \cdot (t + (M - \delta_{-j})n + 1)_{(r+\delta_{-j})n}}{\prod_{j=1}^{s} (t + \delta_j n)_{(M-2\delta_j)n+1}}, \end{aligned}$$

where s, b, r and M are positive integers satisfying certain parity conditions, and δ_{-b}, …, δ_{-1}, δ_1, …, δ_s are nonnegative integers such that $0 \leqslant \delta_{-b} \leqslant \cdots \leqslant \delta_{-1} \leqslant \delta_1 \leqslant \cdots \leqslant \delta_s < M/2$. Then the quantity $S_n = D_{M_1 n}^b \prod_{j=2}^{s} D_{M_j n} \cdot \sum_{\nu=1}^{\infty} R_n^{(b-1)}(\nu)$ can be expressed as a linear form in 1 and odd zeta values with integer coefficients. Here, $D_{M_j n}$ denotes the least common multiple of $1, 2, \dots, M_j n$, and $M_j = \max\{M - 2\delta_1,\ M - \delta_j\}$. The coefficients of the linear form have a large common divisor Φ_n. The presence of Φ_n plays a crucial role in the proof of Zudilin's remarkable theorem [17], which states that at least one of $\zeta(5)$, $\zeta(7)$, $\zeta(9)$, $\zeta(11)$ is irrational. It is worth noting that the irrationality of $\zeta(3)$ was proved by Apéry [1], while the irrationality of $\zeta(5)$ remains an open problem[1].

For a considerable period, it was expected that Zudilin's Φ_n factor would have a negligible impact on asymptotic results as $s \to \infty$. The aim of the current paper is to demonstrate that this previous expectation was inaccurate. In fact, we find that the Φ_n factor arising from the denominator of the rational function does have an asymptotic impact. By incorporating Zudilin's Φ_n factor into the original proofs of the Ball–Rivoal theorem and Sprang's theorem, we establish the following two results, which serve as the main theorems of the current paper:

[1]Note added in revision: a possibly correct proof by Fauzan, dated September 17, 2026, is available. See `https://doi.org/10.5281/zenodo.22826419`.

Theorem 1.3. *For any sufficiently large even integer $s \geqslant s_0$, we have*

$$\dim_{\mathbb{Q}} \operatorname{Span}_{\mathbb{Q}} (1, \zeta(3), \zeta(5), \ldots, \zeta(s-1)) \geqslant \frac{1.009}{1+\log 2} \cdot \log s.$$

Theorem 1.4. *Let p be any prime number. For any sufficiently large even integer $s \geqslant s_0(p)$, we have*

$$\dim_{\mathbb{Q}} \operatorname{Span}_{\mathbb{Q}} (1, \zeta_p(3), \zeta_p(5), \ldots, \zeta_p(s-1)) \geqslant \frac{1.009}{1+\log 2} \cdot \log s.$$

We emphasize that the proofs of Theorems 1.3 and 1.4 do not rely on computer programming. The Φ_n factor has the asymptotic estimate $\Phi_n = \exp(\varpi n + o(n))$ as $n \to \infty$, where ϖ is a computable constant. Typically, determining ϖ requires a substantial amount of computation. For Theorems 1.3 and 1.4, we use very simple parameters to avoid the need for computer assistance. On the other hand, with the help of computer-assisted calculations, we can improve the constant from 1.009 to 1.119 by using complicated parameters. We state the results as follows:

Claim 1.5. *For any sufficiently large even integer $s \geqslant s_0'$, we have*

$$\dim_{\mathbb{Q}} \operatorname{Span}_{\mathbb{Q}} (1, \zeta(3), \zeta(5), \ldots, \zeta(s-1)) \geqslant \frac{1.119356}{1+\log 2} \cdot \log s.$$

For any prime number p, for any sufficiently large even integer $s \geqslant s_0'(p)$, we have

$$\dim_{\mathbb{Q}} \operatorname{Span}_{\mathbb{Q}} (1, \zeta_p(3), \zeta_p(5), \ldots, \zeta_p(s-1)) \geqslant \frac{1.119356}{1+\log 2} \cdot \log s.$$

Note that $1/(1+\log 2) = 0.590\ldots$ and $1.119356/(1+\log 2) = 0.661\ldots$.

Meanwhile, we also obtain the following result as a byproduct:

Claim 1.6. *We have*

$$\dim_{\mathbb{Q}} \operatorname{Span}_{\mathbb{Q}} (1, \zeta(3), \zeta(5), \ldots, \zeta(75)) \geqslant 3.$$

The proofs of Claims 1.5 and 1.6 involve time-consuming calculations (by computer programming). It is unreasonable to require the referee to check the correctness of these time-consuming calculations, as the main ideas of this paper are already conveyed by Theorem 1.3 and Theorem 1.4. Therefore, we state Claims 1.5 and 1.6 as claims rather than theorems. However, we will mention other simple parameters in §13 that improve the constant 1.009 in Theorems 1.3 and 1.4.

We would like to point out that Theorem 1.3 is subsumed by a much stronger result of Fischler, as stated below (but Theorem 1.4 is new):

Theorem 1.7 (Fischler [5], 2026)**.** *For any sufficiently large odd integer s, we have*

$$\dim_{\mathbb{Q}} \operatorname{Span}_{\mathbb{Q}} (1, \zeta(3), \zeta(5), \ldots, \zeta(s)) \geqslant 0.21 \cdot \sqrt{\frac{s}{\log s}}.$$

One notable aspect of Fischler's work is the use of non-explicit rational functions. Because these rational functions are not explicit, proving a certain nonvanishing property becomes challenging. Fischler overcame this difficulty by employing a generalized version of Shidlovsky's lemma. In contrast, our proof of Theorem 1.3 relies on explicit rational functions, making it straightforward to establish the required nonvanishing property. For the p-adic case, our proof of Theorem 1.4 also relies on explicit rational functions. However, the corresponding nonvanishing property is not obvious and requires a more delicate proof.

Although Theorem 1.3 is subsumed by Fischler's result, we present its proof in this paper to highlight the parallels between the classical case and the p-adic case.

1.1 Notation

Throughout this paper, the letters p and q always denote prime numbers. As usual, the symbols $\mathbb{Z}_p$, $\mathbb{Q}_p$, and $\mathbb{C}_p$ denote the ring of p-adic integers, the field of p-adic numbers, and the completion of an algebraic closure of $\mathbb{Q}_p$, respectively. We use $v_p(x)$ to denote the p-adic order of x, normalized by $v_p(p) = 1$. The p-adic norm is defined by $|x|_p = p^{-v_p(x)}$. The notation $|y|$ denotes the Archimedean norm of y. The function $\log(\cdot)$ always denotes the natural logarithm (with respect to the base $e = 2.718\ldots$). The function $\log_p(\cdot)$ denotes the p-adic logarithm, which is defined by

$$\log_p(1+x) := \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} x^k \quad \text{for } x \in \mathbb{C}_p \text{ with } |x|_p < 1.$$

We use the standard asymptotic notations O, o, and $\sim$. For two real-valued functions $f(x)$ and $g(x)$ with $g(x) > 0$, the expressions $f(x) = O(g(x))$, $f(x) = o(g(x))$, and $f(x) \sim g(x)$ mean that $|f(x)| \leqslant Cg(x)$ for some constant $C > 0$, $\lim f(x)/g(x) = 0$, and $\lim f(x)/g(x) = 1$, respectively.

For any positive integer m, we use D_m to denote the least common multiple of $1, 2, \ldots, m$. The Pochhammer symbol $(x)_m$ is defined by

$$(x)_m := x(x+1)\cdots(x+m-1),$$

with the convention $(x)_0 := 1$. The Euler totient function is denoted by $\varphi(\cdot)$. The notation $\phi(\cdot)$ is used to denote some other function in this paper (see (4.6)).

1.2 An outline of the proof of Theorem 1.3

In this subsection, we provide a brief outline of the proof of Theorem 1.3. The reader who is familiar with the work of Ball and Rivoal, and of Zudilin will quickly recognize that our proof follows by incorporating Zudilin's Φ_n factor [17] into the original proof of Ball–Rivoal [2].

Let s be a sufficiently large even integer, and take $r = \lfloor s/\log^2 s \rfloor$. Consider the rational functions:

$$R_n(t) = \frac{(6n)!^{s/2}(4n)!^{s/2}}{n!^{2r}} \cdot (t+3n)$$

$$\times \frac{(t-rn)_{rn}\cdot(t+6n+1)_{rn}}{(t)_{6n+1}^{s/2}\cdot(t+n)_{4n+1}^{s/2}}, \quad n=1,2,3,\ldots.$$

Using arguments similar to those in [2] and [19], we have the following facts:

- The quantity $S_n=\sum_{\nu=1}^{\infty}R_n(\nu)$ can be expressed as a linear form in 1 and odd zeta values with rational coefficients:

$$S_n=\rho_0+\sum_{\substack{3\leqslant i\leqslant s-1\\ i \text{ odd}}}\rho_i\zeta(i).$$

- We have $\Phi_n^{-s/2}D_{6n}^s\cdot\rho_i\in\mathbb{Z}$ for any $i=0,3,5,\ldots,s-1$, provided that $n>s^2$. Here Φ_n is a product of certain primes:

$$\Phi_n=\prod_{\substack{\sqrt{6n}<q\leqslant 6n\\ q \text{ prime}}}q^{\phi(n/q)},$$

  with the function

$$\begin{aligned}\phi(x)&=\inf_{y\in\mathbb{R}}\left(\lfloor 6x\rfloor-\lfloor y\rfloor-\lfloor 6x-y\rfloor+\lfloor 4x\rfloor-\lfloor y-x\rfloor-\lfloor 5x-y\rfloor\right)\\&=\begin{cases}1 & \text{if } \{x\}\in\left[\frac{1}{6},\frac{1}{5}\right)\cup\left[\frac{1}{3},\frac{2}{5}\right)\cup\left[\frac{1}{2},\frac{3}{5}\right)\cup\left[\frac{3}{4},\frac{4}{5}\right),\\ 0 & \text{otherwise.}\end{cases}\end{aligned}$$

  (As usual, $\{x\}$ denotes the fractional part of x.) We remark that Φ_n receives contributions only from the 'denominator type elementary bricks' of the rational function $R_n(t)$.

- We have the following bound for the coefficients ρ_i of the linear forms:

$$\max_{i=0,3,5,\ldots,s-1}|\rho_i|\leqslant\exp\left(\beta(s)n+o(n)\right)\quad\text{as } n\to\infty,$$

  where $\beta(s)$ is a constant depending only on s, and

$$\beta(s)\sim(5\log 2)\cdot s\quad\text{as } s\to\infty.$$

- We have the following asymptotic estimate for the linear forms:

$$S_n=\exp\left(-\alpha(s)n+o(n)\right)\quad\text{as } n\to\infty,$$

  where $\alpha(s)$ is a constant depending only on s, and

$$\alpha(s)\sim 5\cdot s\log s\quad\text{as } s\to\infty.$$

- Let $\psi(x) = \Gamma'(x)/\Gamma(x)$ denote the digamma function. We have the following asymptotic estimate for the Φ_n factor:

$$\Phi_n = \exp\left(\varpi n + o(n)\right) \quad \text{as } n \to \infty,$$

where

$$\begin{aligned}\varpi = &\psi\left(\frac{1}{5}\right) - \psi\left(\frac{1}{6}\right) + \psi\left(\frac{2}{5}\right) - \psi\left(\frac{1}{3}\right) \\ &+ \psi\left(\frac{3}{5}\right) - \psi\left(\frac{1}{2}\right) + \psi\left(\frac{4}{5}\right) - \psi\left(\frac{3}{4}\right) \\ =&2.157479\ldots.\end{aligned}$$

Applying Nesterenko's linear independence criterion to the sequence of linear forms $\{\Phi_n^{-s/2} D_{6n}^s \cdot S_n\}_{n>s^2}$, we obtain

$$\begin{aligned}&\dim_{\mathbb{Q}} \operatorname{Span}_{\mathbb{Q}}\left(1, \zeta(3), \zeta(5), \ldots, \zeta(s-1)\right) \\ \geqslant\ & 1 + \frac{\alpha(s) - 6s + \varpi s/2}{\beta(s) + 6s - \varpi s/2} \\ =\ & (C - o(1)) \cdot \log s \quad \text{as } s \to \infty,\end{aligned}$$

where

$$C = \frac{5}{5\log 2 + 6 - \varpi/2} = \frac{1.009388\ldots}{1 + \log 2}.$$

Therefore, we have $\dim_{\mathbb{Q}} \operatorname{Span}_{\mathbb{Q}}\left(1, \zeta(3), \zeta(5), \ldots, \zeta(s-1)\right) \geqslant (1.009 \cdot \log s)/(1 + \log 2)$ for any sufficiently large even integer s.

To improve the constant from 1.009 to 1.119, we will consider more complicated rational functions $R_n(t)$, as defined in (3.1).

1.3 Similarities between the p-adic case and the classical case

In this subsection, we briefly discuss the similarities (and differences) between the proofs of Theorem 1.3 and Theorem 1.4.

In the classical case, using rational functions to construct linear forms in 1 and zeta values is based on the observation

$$\sum_{t=1}^{\infty} \frac{1}{(t+k)^i} = \zeta(i) - \sum_{\ell=1}^{k} \frac{1}{\ell^i} \quad (k \in \mathbb{Z}_{>0} \text{ and } i \in \mathbb{Z}_{\geqslant 2}). \tag{1.1}$$

We have no straightforward p-adic analogue of (1.1). Instead, we need an intermediate step involving p-adic Hurwitz zeta values. Recall that the classical Hurwitz zeta function is defined by $\zeta(i, x) := \sum_{t=0}^{\infty} (t+x)^{-i}$. We have

$$\sum_{t=0}^{\infty} \frac{1}{(t+k+x)^i} = \zeta(i, x) - \sum_{\ell=0}^{k-1} \frac{1}{(\ell+x)^i} \quad (x \in \mathbb{R}_{>0}).$$

Its p-adic analogue is the following evaluation of the Volkenborn integral:

$$\begin{aligned}&-\int_{\mathbb{Z}_p}\frac{\mathrm{d}t}{(1-i)(t+k+x)^{i-1}}\\&=\omega(x)^{1-i}\zeta_p(i,x)-\sum_{\ell=0}^{k-1}\frac{1}{(\ell+x)^i}\quad(x\in\mathbb{Q}_p\text{ with }|x|_p\geqslant q_p),\end{aligned}$$

where $\omega(x)$ is the Teichmüller character and $\zeta_p(i,x)$ is the p-adic Hurwitz zeta function.

Note that

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{(1-i)(t+k+x)^{i-1}}\right)=\frac{1}{(t+k+x)^i}.$$

It turns out that, for a rational function $R(t)$ of the form

$$R(t)=\sum_{i=1}^{s}\sum_{k=0}^{n}\frac{a_{i,k}}{(t+k)^i}\quad(n\in\mathbb{Z}_{>0},\ s\in\mathbb{Z}_{>0},\text{ and }a_{i,k}\in\mathbb{Q}),$$

we have

$$\sum_{t=0}^{\infty}\frac{\mathrm{d}}{\mathrm{d}t}R(t+x)=\rho_0+\sum_{i=2}^{s+1}\rho_i\zeta(i,x)\quad(x\in\mathbb{R}_{>0}),\tag{1.2}$$

$$-\int_{\mathbb{Z}_p}R(t+x)\,\mathrm{d}t=\rho_0+\sum_{i=2}^{s+1}\rho_i\omega(x)^{1-i}\zeta_p(i,x)\quad(x\in\mathbb{Q}_p\text{ with }|x|_p\geqslant q_p).\tag{1.3}$$

If $x\in\mathbb{Q}_{>0}$ with $|x|_p\geqslant q_p$, then the coefficients $\rho_0,\rho_i\in\mathbb{Q}$ in (1.2) and (1.3) are exactly the same. This is the main similarity between the classical case and the p-adic case.

Equations (1.2) and (1.3) suggest that the p-adic analogue of

$$\sum_{t=0}^{\infty}R(t+x)$$

should be

$$-\int_{\mathbb{Z}_p}W(t+x)\,\mathrm{d}t$$

for some suitable primitive function $W(t)$ (that is, $W'(t)=R(t)$).

However, to prove Theorem 1.3 and Theorem 1.4, we need to use different sequences of rational functions $\{R_n(t)\}_{n\geqslant1}$ and $\{V_n(t)\}_{n\geqslant1}$ (and the primitive functions $\{W_n(t)\}_{n\geqslant1}$, $W_n'(t)=V_n(t)$), respectively. Let us briefly explain the reason. Recall that the original rational functions used by Ball–Rivoal [2] are of the form

$$R_n(t)=(\text{some factor})\cdot\frac{(t-rn)_{rn}(t+n+1)_{rn}}{(t)_{n+1}^s}.\tag{1.4}$$

The 'left factor' $(t-rn)_{rn}$ is designed to ensure that $\sum_{t=1}^{\infty}R_n(t)$ is small. The 'right factor' $(t+n+1)_{rn}$ is designed for symmetry to eliminate all even zeta values. In the

p-adic case, we do not need the symmetry because $\zeta_p(i) = 0$ for any positive even integer i. The candidates for the p-adic case are

$$V_n(t) = (\text{some factor}) \cdot \frac{\prod_{\substack{1 \leqslant \nu \leqslant p^l \\ p \nmid \nu}} (t + \nu/p^l)_n}{(t)_{n+1}^s},$$

where l is a large positive integer. The role of l is similar to that of r in (1.4): it makes

$$-\int_{\mathbb{Z}_p} W_n(t+x)\,\mathrm{d}t \quad \left(x = \frac{\nu}{p^l},\ 1 \leqslant \nu \leqslant p^l,\ p \nmid \nu\right)$$

p-adically small. We need the 'middle factor'

$$\prod_{\substack{1 \leqslant \nu \leqslant p^l \\ p \nmid \nu}} \left(t + \frac{\nu}{p^l}\right)_n$$

of $V_n(t)$ to control the denominator of the coefficient ρ_0 (that is, the coefficient corresponding to 1 in the resulting linear combination of 1 and p-adic odd zeta values). This use of the 'middle factor' also appears in [6, 8, 9]. However, to establish a certain nonvanishing property for the p-adic case, we need a few technical modifications to $V_n(t)$ (see (8.6)).

1.4 Structure of the paper

In Section 2, we introduce Nesterenko's linear independence criterion and its p-adic variant.

Sections 3–6 are devoted to Theorem 1.3. In Section 3, we set up parameters and construct linear forms using explicit rational functions. In Section 4, we prove the arithmetic properties of the linear forms. In Section 5, we study the asymptotic behavior of these linear forms. Finally, in Section 6, we prove Theorem 1.3.

Sections 7–12 are devoted to Theorem 1.4. In Section 7, we recall the definition of p-adic zeta values. In Section 8, we set up parameters and construct linear forms using explicit rational functions. In Section 9, we prove the arithmetic properties of the linear forms. In Section 10, we estimate the Archimedean norm of the coefficients of these linear forms. In Section 11, we estimate the p-adic norm of these linear forms. Finally, in Section 12, we prove Theorem 1.4.

In Section 13, we focus on the computational aspect of our method. We prove Claim 1.5 and Claim 1.6 in this section.

2 Linear independence criteria

Two fundamental criteria for linear independence are Siegel's criterion [15] and Nesterenko's criterion [11]. Each criterion has its own advantages. Fischler's construction [5] is suited to Siegel's criterion, whereas Nesterenko's criterion is convenient for the explicit constructions of Rivoal [14] and Ball–Rivoal [2]. In the current paper, we will utilize Nesterenko's criterion and its p-adic variant to prove Theorem 1.3 and Theorem 1.4, respectively.

Theorem 2.1 (Nesterenko's linear independence criterion [11], 1985)**.** *Let ξ_1, ξ_2, …, ξ_s be real numbers, with $s \in \mathbb{Z}_{>0}$. Let α, β be two positive constants. For any $n \in \mathbb{Z}_{>0}$, let $L_n(X_0, X_1, \dots, X_s) = \sum_{i=0}^{s} l_{n,i} X_i$ be a linear form with integer coefficients $l_{n,i} \in \mathbb{Z}$ for $i = 0, 1, \dots, s$. Suppose the following conditions hold*:

- $|L_n(1, \xi_1, \xi_2, \dots, \xi_s)| = \exp(-\alpha n + o(n))$ *as* $n \to \infty$;
- $\max_{0 \leqslant i \leqslant s} |l_{n,i}| \leqslant \exp(\beta n + o(n))$ *as* $n \to \infty$.

Then, we have

$$\dim_{\mathbb{Q}} \operatorname{Span}_{\mathbb{Q}}(1, \xi_1, \xi_2, \dots, \xi_s) \geqslant 1 + \frac{\alpha}{\beta}.$$

In 2010, Fischler and Zudilin [7] refined Nesterenko's criterion. We present their result below. Although this refined version is not necessary for the proof of Theorem 1.3, it will be utilized in §13 to prove that $\dim_{\mathbb{Q}} \operatorname{Span}_{\mathbb{Q}}(1, \zeta(3), \zeta(5), \dots, \zeta(75)) \geqslant 3$.

Theorem 2.2 (Fischler–Zudilin [7], 2010)**.** *Retain the notation and hypotheses in Theorem 2.1. Suppose, in addition, that for each $n \in \mathbb{Z}_{>0}$ and each $i = 1, 2, \dots, s$, the integer coefficient $l_{n,i}$ has a positive divisor $d_{n,i}$ such that*:

- *$d_{n,i}$ divides $d_{n,i+1}$ for any $n \in \mathbb{Z}_{>0}$ and any $i = 1, 2, \dots, s-1$;*
- *$d_{n,j}/d_{n,i}$ divides $d_{n+1,j}/d_{n+1,i}$ for any $n \in \mathbb{Z}_{>0}$ and any $0 \leqslant i < j \leqslant s$, with $d_{n,0} = 1$;*
- *For each $i = 1, 2, \dots, s$, we have $d_{n,i} = \exp(\gamma_i n + o(n))$ as $n \to \infty$ for some constant $\gamma_i \geqslant 0$.*

Let $d = \dim_{\mathbb{Q}} \operatorname{Span}_{\mathbb{Q}}(1, \xi_1, \xi_2, \dots, \xi_s)$. Then, we have

$$d \geqslant 1 + \frac{\alpha + \gamma_1 + \gamma_2 + \cdots + \gamma_{d-1}}{\beta}.$$

In 2012, Nesterenko [12] proved a p-adic version of his linear independence criterion. We use the following formulation (see [16, Theorem 1.4]):

Theorem 2.3 (Nesterenko's p-adic linear independence criterion [12], 2012)**.** *Let p be any prime number. Let $\xi_1, \xi_2, \dots, \xi_s \in \mathbb{Q}_p$ be p-adic numbers, with $s \in \mathbb{Z}_{>0}$. Let $\alpha_1, \alpha_2, \beta$ be three positive real constants. For any $n \in \mathbb{Z}_{>0}$, let $L_n(X_0, X_1, \dots, X_s) = \sum_{i=0}^{s} l_{n,i} X_i$ be a linear form with integer coefficients $l_{n,i} \in \mathbb{Z}$ for $i = 0, 1, \dots, s$. Suppose the following conditions hold*:

- $\exp(-\alpha_1 n + o(n)) \leqslant |L_n(1, \xi_1, \xi_2, \dots, \xi_s)|_p \leqslant \exp(-\alpha_2 n + o(n))$ *as* $n \to \infty$;
- $\max_{0 \leqslant i \leqslant s} |l_{n,i}| \leqslant \exp(\beta n + o(n))$ *as* $n \to \infty$.

Then, we have

$$\dim_{\mathbb{Q}} \operatorname{Span}_{\mathbb{Q}}(1, \xi_1, \xi_2, \dots, \xi_s) \geqslant \frac{\alpha_1}{\beta + \alpha_1 - \alpha_2}.$$

3 Setup for the proof for the classical case

In this section, we will first set up the parameters. Then, we will construct a sequence of rational functions $R_n(t)$ and utilize them to produce linear forms in 1 and odd zeta values.

3.1 Parameters

Fix a positive integer M and a finite collection of nonnegative integers $\{\delta_j\}_{j=1}^{J}$ such that

$$0 \leqslant \delta_1 \leqslant \delta_2 \leqslant \cdots \leqslant \delta_J < \frac{M}{2},$$

where $J \in \mathbb{Z}_{>0}$ denotes the number of δ_j's.

Let s be a sufficiently large multiple of $2J$. Define the 'numerator length parameter'

$$r := \left\lfloor \frac{s}{\log^2 s} \right\rfloor$$

to be the same as in the Ball–Rivoal setting [2].

3.2 Rational functions

Recall that $(x)_k = x(x+1)\cdots(x+k-1)$ denotes the Pochhammer symbol. For any $n \in \mathbb{Z}_{>0}$, define the following rational function $R_n(t)$:

$$\begin{aligned} R_n(t) :=& \frac{\prod_{j=1}^{J}((M-2\delta_j)n)!^{s/J}}{n!^{2r}} \cdot (2t+Mn) \\ &\times \frac{(t-rn)_{rn}(t+Mn+1)_{rn}}{\prod_{j=1}^{J}(t+\delta_j n)_{(M-2\delta_j)n+1}^{s/J}}. \end{aligned} \tag{3.1}$$

Since s is a multiple of $2J$, this rational function has rational coefficients, that is, $R_n(t) \in \mathbb{Q}(t)$, and it exhibits the following symmetry:

$$R_n(-t-Mn) = -R_n(t). \tag{3.2}$$

Note that the degree of the rational function $R_n(t)$ is

$$\deg R_n = 1 + 2rn - \frac{s}{J}\sum_{j=1}^{J}\left((M-2\delta_j)n+1\right).$$

Since $r = \lfloor s/\log^2 s \rfloor$ and s is assumed to be sufficiently large, we have

$$\deg R_n(t) \leqslant -2 \quad \text{for any } n \in \mathbb{Z}_{>0}. \tag{3.3}$$

Thus, the rational function $R_n(t)$ has the following partial-fraction decomposition:

$$R_n(t) =: \sum_{i=1}^{s}\sum_{k=\delta_1 n}^{(M-\delta_1)n} \frac{a_{n,i,k}}{(t+k)^i}, \tag{3.4}$$

where the coefficients $a_{n,i,k} \in \mathbb{Q}$ (for $1 \leqslant i \leqslant s$ and $\delta_1 n \leqslant k \leqslant (M-\delta_1)n$) are uniquely determined by $R_n(t)$.

3.3 Linear forms

We define

$$\rho_{n,i} := \sum_{k=\delta_1 n}^{(M-\delta_1)n} a_{n,i,k} \quad (i = 1, 2, 3, \dots, s), \tag{3.5}$$

$$\rho_{n,0} := -\sum_{i=1}^{s} \sum_{k=\delta_1 n}^{(M-\delta_1)n} \sum_{\ell=1}^{k} \frac{a_{n,i,k}}{\ell^i}. \tag{3.6}$$

Note that the symmetry of the rational function (Equation (3.2)) implies the symmetry of its coefficients in the partial-fraction decomposition; that is

$$a_{n,i,k} = (-1)^{i+1} a_{n,i,Mn-k} \quad (1 \leqslant i \leqslant s \text{ and } \delta_1 n \leqslant k \leqslant (M-\delta_1)n).$$

Therefore, we have

$$\rho_{n,i} = 0 \quad \text{for } i = 2, 4, 6, \dots, s.$$

On the other hand, by (3.3) and (3.4), we have

$$\rho_{n,1} = \lim_{t\to\infty} tR_n(t) = 0.$$

Define

$$S_n := \sum_{\nu=1}^{\infty} R_n(\nu). \tag{3.7}$$

Using arguments similar to those in [2, Lemme 1], we have the following lemma:

Lemma 3.1. *For any $n \in \mathbb{Z}_{>0}$, S_n is a linear form in 1 and odd zeta values, given by*

$$S_n = \rho_{n,0} + \sum_{\substack{3\leqslant i\leqslant s-1 \\ i \text{ odd}}} \rho_{n,i}\zeta(i),$$

where the coefficients $\rho_{n,i}$ (for $i = 3, 5, \dots, s-1$) are defined by (3.5), and $\rho_{n,0}$ is defined by (3.6).

4 Arithmetic properties for the classical case

In this section, we will first introduce two types of 'elementary bricks': the 'denominator type elementary brick' (see Lemma 4.1) and the 'numerator type elementary brick' (see Lemma 4.2). Our rational functions in (3.1) and many other rational functions in the literature (for example, [14, 17]) are finite products of these 'elementary bricks'. Then, we will study the denominators of the coefficients $a_{n,i,k}$ (defined by (3.4)) and $\rho_{n,i}$ (defined by (3.6) and (3.5)).

Recall that we denote by

$$D_m = \operatorname{lcm}[1, 2, 3, \dots, m]$$

the least common multiple of the numbers $1, 2, 3, \dots, m$ for any positive integer m. We denote by $f^{(\lambda)}(t)$ the λth derivative of a function $f(t)$ for any nonnegative integer λ. For a prime number p, the notation $v_p(x)$ denotes the p-adic order of x.

Lemma 4.1. *Let a, b, a_0, b_0 be integers such that $a_0 \leqslant a \leqslant b \leqslant b_0$ and $b_0 > a_0$. Consider the 'denominator type elementary brick'*

$$G(t) = \frac{(b-a)!}{(t+a)_{b-a+1}}.$$

Then, we have

$$D^{\lambda}_{b_0-a_0} \cdot \frac{1}{\lambda!} \left(G(t)(t+k)\right)^{(\lambda)} \big|_{t=-k} \in \mathbb{Z} \tag{4.1}$$

for any $k \in [a_0, b_0] \cap \mathbb{Z}$ and any $\lambda \in \mathbb{Z}_{\geqslant 0}$.

Moreover, for any prime number $q > \sqrt{b_0 - a_0}$, any $k \in [a_0, b_0] \cap \mathbb{Z}$, and any $\lambda \in \mathbb{Z}_{\geqslant 0}$, we have

$$v_q\left(\left(G(t)(t+k)\right)^{(\lambda)} \big|_{t=-k}\right) \geqslant -\lambda + \left\lfloor \frac{b-a}{q} \right\rfloor - \left\lfloor \frac{k-a}{q} \right\rfloor - \left\lfloor \frac{b-k}{q} \right\rfloor. \tag{4.2}$$

Proof. See [19, Lemmas 16 and 18]. We have replaced b and b_0 in [19, Lemmas 16 and 18] by $b+1$ and b_0+1, respectively. □

Lemma 4.2. *Let a, b, c, m be integers with $m > 0$ and $b > 0$. Consider the 'numerator type elementary brick'*

$$F(t) = b^m \prod_{\substack{q \mid b \\ q \text{ prime}}} q^{\lfloor m/(q-1) \rfloor} \cdot \frac{(ct + a/b)_m}{m!}.$$

Then, we have

$$D^{\lambda}_{m} \cdot \frac{1}{\lambda!} F^{(\lambda)}(t) \big|_{t=-k} \in \mathbb{Z}$$

for any $k \in \mathbb{Z}$ and any $\lambda \in \mathbb{Z}_{\geqslant 0}$.

Lemma 4.2 is known in the literature but not stated as above. For example, one can slightly modify the arguments in [9, Proposition 3.2] to obtain a proof. For completeness, we give the proof.

Proof of Lemma 4.2. Fix $k \in \mathbb{Z}$ and $\lambda \in \mathbb{Z}_{\geqslant 0}$. Note that

$$F(t-k) = \frac{\prod_{q \text{ prime}} q^{\lfloor m/(q-1) \rfloor}}{m!} \cdot \prod_{\nu=0}^{m-1} (bct - bck + b\nu + a).$$

Write

$$\prod_{\nu=0}^{m-1} (bct - bck + b\nu + a) = \sum_{j=0}^{m} C_j t^j, \quad \text{where } C_j \in \mathbb{Z} \text{ for } j = 0, 1, \ldots, m. \tag{4.3}$$

Then, we have

$$\frac{1}{\lambda!} F^{(\lambda)}(t) \big|_{t=-k} = \frac{\prod_{q \text{ prime}} q^{\lfloor m/(q-1) \rfloor}}{m!} \cdot C_\lambda,$$

with the convention $C_\lambda = 0$ for $\lambda > m$.

It remains to prove that, for any prime q_0, we have

$$v_{q_0}\left(D_m^\lambda \frac{\prod_{q \text{ prime}} q^{\lfloor m/(q-1)\rfloor}}{m!} C_\lambda\right) \geqslant 0. \tag{4.4}$$

Since $C_\lambda \in \mathbb{Z}$ and $v_{q_0}(m!) \leqslant \lfloor m/(q_0-1)\rfloor$, Inequality (4.4) holds for any prime $q_0 \mid b$. Now, assume $q_0 \nmid b$. By expanding the left-hand side of (4.3), we see that C_λ can be expressed as a sum of finitely many terms of the form

$$(bc)^\lambda \prod_{i=1}^{\lambda+1} U_i,$$

where each U_i is a product of $\ell_i \in \mathbb{Z}_{\geqslant 0}$ consecutive terms in an integer arithmetic progression with common difference b such that $\sum_{i=1}^{\lambda+1} \ell_i = m - \lambda$. Since $q_0 \nmid b$, we have

$$v_{q_0}(U_i) \geqslant \sum_{j=1}^{\infty} \left\lfloor \frac{\ell_i}{q_0^j} \right\rfloor,$$

and hence

$$v_{q_0}\left((bc)^\lambda \prod_{i=1}^{\lambda+1} U_i\right) \geqslant \sum_{j=1}^{\infty} \sum_{i=1}^{\lambda+1} \left\lfloor \frac{\ell_i}{q_0^j} \right\rfloor.$$

Noting that

$$\sum_{i=1}^{\lambda+1} \left\lfloor \frac{\ell_i}{q_0^j} \right\rfloor \geqslant \sum_{i=1}^{\lambda+1} \frac{\ell_i - (q_0^j - 1)}{q_0^j} = \frac{m+1}{q_0^j} - \lambda - 1 > \left\lfloor \frac{m}{q_0^j} \right\rfloor - \lambda - 1$$

and the left-hand side above is an integer, we have

$$\sum_{i=1}^{\lambda+1} \left\lfloor \frac{\ell_i}{q_0^j} \right\rfloor \geqslant \left\lfloor \frac{m}{q_0^j} \right\rfloor - \lambda.$$

Thus,

$$\begin{aligned} v_{q_0}\left((bc)^\lambda \prod_{i=1}^{\lambda+1} U_i\right) &\geqslant \sum_{j=1}^{\infty} \sum_{i=1}^{\lambda+1} \left\lfloor \frac{\ell_i}{q_0^j} \right\rfloor = \sum_{j=1}^{\lfloor \log m/\log q_0 \rfloor} \sum_{i=1}^{\lambda+1} \left\lfloor \frac{\ell_i}{q_0^j} \right\rfloor \\ &\geqslant \sum_{j=1}^{\lfloor \log m/\log q_0 \rfloor} \left(\left\lfloor \frac{m}{q_0^j} \right\rfloor - \lambda \right) = v_{q_0}(m!) - \lambda v_{q_0}(D_m). \end{aligned}$$

Therefore, we have $v_{q_0}(C_\lambda) \geqslant v_{q_0}(m!) - \lambda v_{q_0}(D_m)$, which proves Inequality (4.4). □

Now, we study the arithmetic property of the coefficients $a_{n,i,k}$ defined by (3.4). The proof of the following lemma is essentially the same as (part of) the proof of [19, Lemma 19], but we would like to present the details here.

Lemma 4.3. *For any integer $n > s^2$, we have*

$$\Phi_n^{-s/J} D_{(M-2\delta_1)n}^{s-i} \cdot a_{n,i,k} \in \mathbb{Z} \quad (1 \leqslant i \leqslant s \text{ and } \delta_1 n \leqslant k \leqslant (M-\delta_1)n),$$

where the factor Φ_n is a product of certain primes:

$$\Phi_n := \prod_{\substack{\sqrt{Mn} < q \leqslant (M-2\delta_1)n \\ q \text{ prime}}} q^{\phi(n/q)}, \tag{4.5}$$

and the function $\phi(\cdot)$ is defined by

$$\phi(x) := \inf_{y \in \mathbb{R}} \sum_{j=1}^{J} \left(\lfloor (M-2\delta_j)x \rfloor - \lfloor y - \delta_j x \rfloor - \lfloor (M-\delta_j)x - y \rfloor \right). \tag{4.6}$$

Proof. Fix any $i \in \{1, 2, \ldots, s\}$ and any $k \in [\delta_1 n,\ (M-\delta_1)n] \cap \mathbb{Z}$.

By (3.4), we have

$$a_{n,i,k} = \frac{1}{(s-i)!} \left(R_n(t)(t+k)^s \right)^{(s-i)} \Big|_{t=-k}. \tag{4.7}$$

Consider the following elementary bricks:

$$\begin{aligned}
F_0(t) &= 2t + Mn, \\
F_{-,\tau}(t) &= \frac{(t-(r-\tau)n)_n}{n!}, \quad \tau = 0, 1, 2, \ldots, r-1, \\
F_{+,\tau}(t) &= \frac{(t+(M+\tau)n+1)_n}{n!}, \quad \tau = 0, 1, 2, \ldots, r-1, \\
G_j(t) &= \frac{((M-2\delta_j)n)!}{(t+\delta_j n)_{(M-2\delta_j)n+1}}, \quad j = 1, 2, \ldots, J.
\end{aligned}$$

From the definition of the rational function $R_n(t)$ (see Equation (3.1)), we have

$$R_n(t)(t+k)^s = F_0(t) \cdot \prod_{\tau=0}^{r-1} F_{-,\tau}(t) F_{+,\tau}(t) \cdot \prod_{j=1}^{J} \left(G_j(t)(t+k) \right)^{s/J}. \tag{4.8}$$

By Lemma 4.2, for any polynomial $F(t)$ of the form $F_0(t)$, $F_{-,\tau}(t)$, $F_{+,\tau}(t)$, we have

$$D_n^{\lambda} \cdot \frac{1}{\lambda!} F^{(\lambda)}(t) \big|_{t=-k} \in \mathbb{Z} \tag{4.9}$$

for any $\lambda \in \mathbb{Z}_{\geqslant 0}$. By (4.1) of Lemma 4.1 (with $a_0 = \delta_1 n$ and $b_0 = (M-\delta_1)n$), we have

$$D_{(M-2\delta_1)n}^{\lambda} \cdot \frac{1}{\lambda!} \left(G_j(t)(t+k) \right)^{(\lambda)} \big|_{t=-k} \in \mathbb{Z} \tag{4.10}$$

for any $j = 1, 2, \ldots, J$ and any $\lambda \in \mathbb{Z}_{\geqslant 0}$. Now, substituting (4.8) into (4.7), applying the Leibniz rule, and using (4.9) and (4.10), we obtain

$$D_{(M-2\delta_1)n}^{s-i} \cdot a_{n,i,k} \in \mathbb{Z}. \tag{4.11}$$

Moreover, for any prime q such that $\sqrt{Mn} < q \leqslant (M-2\delta_1)n$, taking also (4.2) of Lemma 4.1 into consideration, and noting that $q > \sqrt{Mn} > s$ (because $n > s^2$), we have

$$\begin{aligned} v_q\left(a_{n,i,k}\right) &= v_q\left(\left(R_n(t)(t+k)^s\right)^{(s-i)}\Big|_{t=-k}\right) \\ &\geqslant -(s-i) + \frac{s}{J}\cdot\sum_{j=1}^{J}\left(\left\lfloor\frac{(M-2\delta_j)n}{q}\right\rfloor - \left\lfloor\frac{k-\delta_j n}{q}\right\rfloor - \left\lfloor\frac{(M-\delta_j)n-k}{q}\right\rfloor\right) \\ &\geqslant -(s-i) + \frac{s}{J}\cdot\phi\left(\frac{n}{q}\right), \end{aligned} \tag{4.12}$$

where the function $\phi(\cdot)$ is given by (4.6). Combining (4.11) and (4.12), we obtain the desired conclusion

$$\Phi_n^{-s/J} D^{s-i}_{(M-2\delta_1)n}\cdot a_{n,i,k} \in \mathbb{Z}. \qquad \square$$

The arithmetic property of $a_{n,i,k}$ propagates to the arithmetic property of $\rho_{n,i}$ (defined by (3.5) and (3.6)), as the following lemma shows.

Lemma 4.4. *For any integer $n > s^2$, we have*

$$\Phi_n^{-s/J} D^{s-i}_{(M-2\delta_1)n}\cdot \rho_{n,i} \in \mathbb{Z} \quad (i = 1, 2, \ldots, s),$$

and

$$\Phi_n^{-s/J}\prod_{j=1}^{J} D^{s/J}_{M_j n}\cdot \rho_{n,0} \in \mathbb{Z},$$

where

$$M_j = \max\{M-2\delta_1,\ M-\delta_j\}.$$

Proof. The first assertion follows directly from (3.5) and Lemma 4.3.

Now, using (3.6), we have

$$\rho_{n,0} = -\sum_{i=1}^{s}\sum_{k=\delta_1 n}^{(M-\delta_1)n} a_{n,i,k}\sum_{\ell=1}^{k}\frac{1}{\ell^i}. \tag{4.13}$$

Fix any $i \in \{1, 2, \ldots, s\}$ and let

$$j^* = \left\lceil\frac{iJ}{s}\right\rceil.$$

Note that we have $a_{n,i,k} = 0$ unless $\delta_{j^*} n \leqslant k \leqslant (M-\delta_{j^*})n$ (by examining the order of the pole at $t = -k$ in (3.1)). Thus,

$$\sum_{k=\delta_1 n}^{(M-\delta_1)n} a_{n,i,k}\sum_{\ell=1}^{k}\frac{1}{\ell^i} = \sum_{k=\delta_{j^*} n}^{(M-\delta_{j^*})n} a_{n,i,k}\sum_{\ell=1}^{k}\frac{1}{\ell^i}. \tag{4.14}$$

Clearly,

$$D^i_{(M-\delta_{j^*})n}\cdot\sum_{\ell=1}^{k}\frac{1}{\ell^i} \in \mathbb{Z} \quad \text{for any } k \leqslant (M-\delta_{j^*})n.$$

On the other hand, Lemma 4.3 implies

$$\Phi_n^{-s/J} D_{(M-2\delta_1)n}^{s-i} \cdot a_{n,i,k} \in \mathbb{Z}.$$

Noting that $D_{(M-\delta_{j^*})n}^{i} D_{(M-2\delta_1)n}^{s-i}$ is a divisor of $\prod_{j=1}^{J} D_{M_j n}^{s/J}$, we obtain

$$\Phi_n^{-s/J} \prod_{j=1}^{J} D_{M_j n}^{s/J} \cdot \sum_{k=\delta_{j^*} n}^{(M-\delta_{j^*})n} a_{n,i,k} \sum_{\ell=1}^{k} \frac{1}{\ell^i} \in \mathbb{Z}. \tag{4.15}$$

Combining (4.13), (4.14), and (4.15), we conclude that

$$\Phi_n^{-s/J} \prod_{j=1}^{J} D_{M_j n}^{s/J} \cdot \rho_{n,0} \in \mathbb{Z}. \qquad \square$$

5 Asymptotic estimates for the classical case

In this section, we will study the asymptotic behavior of the linear form S_n (defined by (3.7)), the coefficients $\rho_{n,i}$ (defined by (3.5) and (3.6)), and the factor Φ_n (defined by (4.5)) as $n \to \infty$.

Lemma 5.1. *We have*

$$S_n = \exp\left(-\alpha(s)n + o(n)\right) \quad \text{as } n \to \infty, \tag{5.1}$$

where $\alpha(s)$ is a constant independent of n. Moreover, we have

$$\alpha(s) \sim \frac{s \log s}{J} \sum_{j=1}^{J} (M - 2\delta_j) \quad \text{as } s \to \infty. \tag{5.2}$$

Proof. Consider the Lebesgue integral

$$I_n = \int_{[0,1]^{s+1}} F(x_0, x_1, \ldots, x_s)^n G(x_0, x_1, \ldots, x_s)\, \mathrm{d}x_0 \mathrm{d}x_1 \cdots \mathrm{d}x_s, \tag{5.3}$$

where

$$\begin{aligned} &F(x_0, x_1, \ldots, x_s) \\ &= \frac{x_0^r (1-x_0)^M \prod_{j=1}^{J} \prod_{k=1}^{s/J} x_{(j-1)s/J+k}^{r+\delta_j} (1 - x_{(j-1)s/J+k})^{M-2\delta_j}}{(1 - x_0 x_1 \cdots x_s)^{2r+M}} \end{aligned} \tag{5.4}$$

for $(x_0, x_1, \ldots, x_s) \in [0,1]^{s+1} \setminus \{(1, 1, \ldots, 1)\}$ with $F(1, 1, \ldots, 1) = 0$, and

$$G(x_0, x_1, \ldots, x_s) = \frac{1 + x_0 x_1 \cdots x_s}{(1 - x_0 x_1 \cdots x_s)^3}$$

for $(x_0, x_1, \dots, x_s) \in [0,1]^{s+1} \setminus \{(1,1,\dots,1)\}$ with $G(1,1,\dots,1) = 0$.

We claim that

$$S_n = \frac{((2r+M)n+2)!}{(Mn)! \cdot n!^{2r}} I_n. \tag{5.5}$$

In fact, using the expansion

$$\begin{aligned}
&\frac{1 + x_0 x_1 \dots x_s}{(1 - x_0 x_1 \cdots x_s)^{(2r+M)n+3}} \\
=&\frac{1}{((2r+M)n+2)!} \sum_{\nu=1}^{\infty} (\nu)_{(2r+M)n+1} \cdot ((2r+M)n+2\nu) \cdot x_0^{\nu-1} x_1^{\nu-1} \cdots x_s^{\nu-1}
\end{aligned}$$

and the well-known identity for Euler's Beta function

$$\int_0^1 x^{(r+\delta)n+\nu-1} (1-x)^{(M-2\delta)n} \, \mathrm{d}x = \frac{((M-2\delta)n)!}{((r+\delta)n+\nu)_{(M-2\delta)n+1}},$$

we deduce from Beppo Levi's lemma and Tonelli's theorem that

$$\begin{aligned}
I_n &= \frac{(Mn)!}{((2r+M)n+2)!} \\
&\quad \times \sum_{\nu=1}^{\infty} \frac{(\nu)_{(2r+M)n+1} \cdot ((2r+M)n+2\nu)}{(rn+\nu)_{Mn+1}} \prod_{j=1}^{J} \frac{((M-2\delta_j)n)!^{s/J}}{((r+\delta_j)n+\nu)^{s/J}_{(M-2\delta_j)n+1}} \\
&= \frac{(Mn)! \cdot n!^{2r}}{((2r+M)n+2)!} \sum_{\nu=1}^{\infty} R_n(rn+\nu).
\end{aligned}$$

Then, by (3.7) and the fact that $R_n(\nu) = 0$ for $\nu = 1, 2, 3, \dots, rn$, we conclude that (5.5) holds.

It is easy to see that $G(x_0, x_1, \dots, x_s)$ is Lebesgue integrable over $[0,1]^{s+1}$. Also, $G(x_0, x_1, \dots, x_s)$ is positive and continuous over $[0,1]^{s+1} \setminus \{(1,1,\dots,1)\}$. We claim that $F(x_0, x_1, \dots, x_s)$ is continuous on $[0,1]^{s+1}$. In fact, the only possible discontinuity of $F(x_0, x_1, \dots, x_s)$ is at $(1,1,\dots,1)$. Since $1 - x_0 x_1 \dots x_s \geqslant 1 - x_i$ for every $i = 0, 1, \dots, s$ on $[0,1]^{s+1}$, we have

$$(1 - x_0 x_1 \cdots x_s)^{s+1} \geqslant \prod_{i=0}^{s} (1 - x_i),$$

and hence

$$\begin{aligned}
&F(x_0, x_1, \dots, x_s) \\
&\leqslant x_0^r (1-x_0)^{M - \frac{2r+M}{s+1}} \prod_{j=1}^{J} \prod_{k=1}^{s/J} x_{(j-1)s/J+k}^{r+\delta_j} (1 - x_{(j-1)s/J+k})^{M - 2\delta_j - \frac{2r+M}{s+1}}
\end{aligned} \tag{5.6}$$

for all $(x_0, x_1, \dots, x_s) \in [0,1]^{s+1} \setminus \{(1,1,\dots,1)\}$. Since $r = \lfloor s/\log^2 s \rfloor$ and s is sufficiently large (so that $(2r+M)/(s+1) < 1$), inequality (5.6) implies the continuity of $F(x_0, x_1, \dots, x_s)$ at $(1,1,\dots,1)$. Thus, by (5.3), we have

$$\lim_{n\to\infty} I_n^{1/n} = \max_{(x_0, x_1, \dots, x_s) \in [0,1]^{s+1}} F(x_0, x_1, \dots, x_s).$$

Then, by (5.5) and Stirling's formula, we have

$$S_n = \exp\left(-\alpha(s)n + o(n)\right) \quad \text{as } n \to \infty,$$

where

$$\begin{aligned}\alpha(s) = {} & M\log M - (2r+M)\log(2r+M) \\ & - \log \max_{(x_0,x_1,\dots,x_s)\in[0,1]^{s+1}} F(x_0,x_1,\dots,x_s).\end{aligned} \tag{5.7}$$

Finally, by (5.7) and $r = \lfloor s/\log^2 s \rfloor$, to prove the last assertion about the asymptotic behavior of $\alpha(s)$ as $s \to \infty$, it remains to prove that

$$-\log \max_{(x_0,x_1,\dots,x_s)\in[0,1]^{s+1}} F(x_0,x_1,\dots,x_s) \sim \frac{s\log s}{J}\sum_{j=1}^{J}(M-2\delta_j) \quad \text{as } s\to\infty. \tag{5.8}$$

In fact, taking $x_0 = x_1 = \cdots = x_s = 1 - 1/r$ in (5.4), we have (recall $r = \lfloor s/\log^2 s \rfloor$)

$$\begin{aligned}& \log \max_{(x_0,x_1,\dots,x_s)\in[0,1]^{s+1}} F(x_0,x_1,\dots,x_s) \\ \geqslant {} & \left((s+1)r + \frac{s}{J}\sum_{j=1}^{J}\delta_j\right)\log\left(1-\frac{1}{r}\right) - \left(M + \frac{s}{J}\sum_{j=1}^{J}(M-2\delta_j)\right)\log r \\ & - (2r+M)\log\left(1-\left(1-\frac{1}{r}\right)^{s+1}\right) \\ = {} & -\frac{s\log s}{J}\sum_{j=1}^{J}(M-2\delta_j) - O(s\log\log s) \quad \text{as } s\to\infty.\end{aligned} \tag{5.9}$$

On the other hand, applying the elementary inequality

$$x^a(1-x)^b \leqslant \frac{a^a b^b}{(a+b)^{a+b}} \quad (\text{for any } a>0,\ b>0,\ \text{and } x\in[0,1])$$

to (5.6), we have

$$\begin{aligned}& \log \max_{(x_0,x_1,\dots,x_s)\in[0,1]^{s+1}} F(x_0,x_1,\dots,x_s) \\ \leqslant {} & r\log r + \left(M - \frac{2r+M}{s+1}\right)\log\left(M-\frac{2r+M}{s+1}\right) \\ & - \left(r+M-\frac{2r+M}{s+1}\right)\log\left(r+M-\frac{2r+M}{s+1}\right) \\ & + \frac{s}{J}\sum_{j=1}^{J}(r+\delta_j)\log(r+\delta_j) \\ & + \frac{s}{J}\sum_{j=1}^{J}\left(M-2\delta_j-\frac{2r+M}{s+1}\right)\log\left(M-2\delta_j-\frac{2r+M}{s+1}\right)\end{aligned}$$

$$
\begin{aligned}
&- \frac{s}{J}\sum_{j=1}^{J}\left(r+M-\delta_j-\frac{2r+M}{s+1}\right)\log\left(r+M-\delta_j-\frac{2r+M}{s+1}\right)\\
&=\frac{s}{J}\sum_{j=1}^{J}(r+\delta_j)\log r-\frac{s}{J}\sum_{j=1}^{J}(r+M-\delta_j)\log r+O(s)\\
&=-\frac{s\log s}{J}\sum_{j=1}^{J}(M-2\delta_j)+O(s\log\log s)\quad \text{as } s\to\infty. \qquad (5.10)
\end{aligned}
$$

Combining (5.9) and (5.10), we conclude that (5.8) holds. The proof of Lemma 5.1 is complete. □

Recall that $\rho_{n,i}$ $(i=0,1,2,\dots,s)$ are defined by (3.6) and (3.5).

Lemma 5.2. *We have*

$$
\max_{0\leqslant i\leqslant s}|\rho_{n,i}|\leqslant \exp\left(\beta(s)n+o(n)\right)\quad \text{as } n\to\infty, \qquad (5.11)
$$

where $\beta(s)$ is a constant independent of n. Moreover, we have

$$
\beta(s)\sim\log 2\cdot\frac{s}{J}\sum_{j=1}^{J}(M-2\delta_j)\quad \text{as } s\to\infty. \qquad (5.12)
$$

Proof. By (3.6) and (3.5), we have

$$
\max_{0\leqslant i\leqslant s}|\rho_{n,i}|\leqslant (Mn+1)^2 s\cdot\max_{i,k}|a_{n,i,k}|, \qquad (5.13)
$$

where $\max_{i,k}$ is taken over all $i\in\{1,2,\dots,s\}$ and $k\in[\delta_1 n,(M-\delta_1)n]\cap\mathbb{Z}$.

Now fix any pair (i,k) in the range above. By (3.4) and Cauchy's integral formula, we have

$$
a_{n,i,k}=\frac{1}{2\pi\sqrt{-1}}\int_{|z+k|=1/10}(z+k)^{i-1}R_n(z)\,\mathrm{d}z.
$$

Therefore, we have (recall (3.1))

$$
\begin{aligned}
|a_{n,i,k}|&\leqslant\max_{|z+k|=1/10}|R_n(z)|\\
&=\frac{\prod_{j=1}^{J}((M-2\delta_j)n)!^{s/J}}{n!^{2r}}\\
&\quad\times\max_{|z+k|=1/10}|2z+Mn|\frac{|(z-rn)_{rn}(z+Mn+1)_{rn}|}{\prod_{j=1}^{J}|(z+\delta_j n)_{(M-2\delta_j)n+1}|^{s/J}}. \qquad (5.14)
\end{aligned}
$$

In the following, the complex number z is always assumed to be on the circle $|z+k|=1/10$. We will use the triangle inequality to bound (5.14) from above.

Obviously, we have

$$
\max_{|z+k|=1/10}|2z+Mn|\leqslant 4Mn. \qquad (5.15)
$$

For any $\nu \in \{1, 2, \ldots, rn\}$, we have

$$|(z-\nu)(z+Mn+\nu)| \leqslant (k+\nu+1)(Mn-k+\nu+1) \leqslant \left(\frac{Mn}{2}+\nu+1\right)^2.$$

Therefore,

$$\begin{aligned}
&\max_{|z+k|=1/10} |(z-rn)_{rn}(z+Mn+1)_{rn}| \\
=\ &\max_{|z+k|=1/10} \prod_{\nu=1}^{rn} |(z-\nu)(z+Mn+\nu)| \\
\leqslant\ &\prod_{\nu=1}^{rn} \left(\left\lceil \frac{Mn}{2} \right\rceil + \nu + 1\right)^2 \\
=\ &\frac{\left(\lceil \frac{Mn}{2} \rceil + rn + 1\right)!^2}{\left(\lceil \frac{Mn}{2} \rceil + 1\right)!^2}.
\end{aligned} \tag{5.16}$$

Now, if $\delta_j n < k < (M-\delta_j)n$, then

$$\begin{aligned}
|(z+\delta_j n)_{(M-2\delta_j)n+1}| &\geqslant (k-\delta_j n-1)! \cdot \frac{9}{10} \cdot \frac{1}{10} \cdot \frac{9}{10} \cdot ((M-\delta_j)n-k-1)! \\
&\geqslant \frac{1}{10^3 M^2 n^2} \cdot (k-\delta_j n)! \cdot ((M-\delta_j)n-k)!,
\end{aligned}$$

and hence

$$\frac{((M-2\delta_j)n)!}{|(z+\delta_j n)_{(M-2\delta_j)n+1}|} \leqslant 10^3 M^2 n^2 \binom{(M-2\delta_j)n}{k-\delta_j n} \leqslant 10^3 M^2 n^2 \cdot 2^{(M-2\delta_j)n}.$$

If $k \leqslant \delta_j n$, then

$$\begin{aligned}
|(z+\delta_j n)_{(M-2\delta_j)n+1}| &\geqslant \frac{1}{10} \cdot \frac{9}{10} \cdot ((M-2\delta_j)n-1)! \\
&\geqslant \frac{1}{10^2 Mn}((M-2\delta_j)n)!,
\end{aligned}$$

and hence

$$\frac{((M-2\delta_j)n)!}{|(z+\delta_j n)_{(M-2\delta_j)n+1}|} \leqslant 10^2 Mn.$$

Similarly, if $k \geqslant (M-\delta_j)n$, we also have

$$\frac{((M-2\delta_j)n)!}{|(z+\delta_j n)_{(M-2\delta_j)n+1}|} \leqslant 10^2 Mn.$$

In any case, we always have

$$\max_{|z+k|=1/10} \frac{((M-2\delta_j)n)!}{|(z+\delta_j n)_{(M-2\delta_j)n+1}|} \leqslant 10^3 M^2 n^2 \cdot 2^{(M-2\delta_j)n}. \tag{5.17}$$

Combining inequalities (5.13)–(5.17) and using Stirling's formula, we have

$$\max_{0\leqslant i\leqslant s}|\rho_{n,i}|\leqslant \exp\left(\beta(s)n+o(n)\right)\quad \text{as } n\to\infty,$$

where

$$\beta(s)=(2r+M)\log\left(r+\frac{M}{2}\right)-M\log\frac{M}{2}+\log 2\cdot\frac{s}{J}\sum_{j=1}^{J}(M-2\delta_j).$$

Since $r=\lfloor s/\log^2 s\rfloor$, we have

$$\beta(s)\sim\log 2\cdot\frac{s}{J}\sum_{j=1}^{J}(M-2\delta_j)\quad \text{as } s\to\infty. \qquad \square$$

At the end of this section, we estimate the factor Φ_n (defined by (4.5)). It is well known that the prime number theorem implies

$$D_m=e^{m+o(m)}\quad \text{as } m\to\infty. \tag{5.18}$$

The following lemma is also a corollary of the prime number theorem.

Lemma 5.3 (cf. [18, Lemma 4.4])**.** *For the factor Φ_n defined by* (4.5)*, we have*

$$\Phi_n=\exp\left(\varpi n+o(n)\right)\quad \textit{as } n\to\infty,$$

where the constant ϖ is given by

$$\varpi=\int_0^1\phi(x)\,\mathrm{d}\psi(x)-\int_0^{1/(M-2\delta_1)}\phi(x)\,\frac{\mathrm{d}x}{x^2}. \tag{5.19}$$

Here, the function $\psi(x)=\Gamma'(x)/\Gamma(x)$ is the digamma function, and the function $\phi(x)$ is defined by (4.6)*.*

6 Proof of Theorem 1.3

In this section, we will first reduce the proof of Theorem 1.3 to a computational task. Then, we will prove Theorem 1.3 using some very simple parameters.

Proposition 6.1. *Let $\psi(\cdot)$ be the digamma function. For any collection of nonnegative integers $(M,\delta_1,\delta_2,\dots,\delta_J)$ under the constraints $J\geqslant 1$ and*

$$0\leqslant\delta_1\leqslant\delta_2\leqslant\cdots\leqslant\delta_J<\frac{M}{2},$$

we define

$$\varpi=\int_0^1\phi(x)\,\mathrm{d}\psi(x)-\int_0^{1/(M-2\delta_1)}\phi(x)\,\frac{\mathrm{d}x}{x^2},$$

where the function $\phi(\cdot)$ is given by

$$\phi(x) = \inf_{y\in\mathbb{R}} \sum_{j=1}^{J} \left(\lfloor (M-2\delta_j)x\rfloor - \lfloor y-\delta_j x\rfloor - \lfloor (M-\delta_j)x - y\rfloor\right).$$

Then, we have

$$\dim_{\mathbb{Q}} \mathrm{Span}_{\mathbb{Q}}\left(1, \zeta(3), \zeta(5), \ldots, \zeta(s-1)\right) \geqslant (C - o(1)) \cdot \log s$$

as the positive even integer $s \to \infty$, where the constant C is defined by

$$C = \frac{\sum_{j=1}^{J}(M-2\delta_j)}{\log 2 \cdot \sum_{j=1}^{J}(M-2\delta_j) - \varpi + \sum_{j=1}^{J} \max\{M-2\delta_1,\ M-\delta_j\}}. \tag{6.1}$$

Proof. For any integer $n > s^2$, consider

$$\widehat{S}_n := \Phi_n^{-s/J} \prod_{j=1}^{J} D_{M_j n}^{s/J} \cdot S_n,$$

where $M_j = \max\{M-2\delta_1,\ M-\delta_j\}$. By Lemmas 3.1 and 4.4,

$$\widehat{S}_n = \widehat{\rho}_{n,0} + \sum_{\substack{3\leqslant i\leqslant s-1 \\ i \text{ odd}}} \widehat{\rho}_{n,i}\zeta(i)$$

is a linear form in 1 and odd zeta values with integer coefficients:

$$\widehat{\rho}_{n,i} := \Phi_n^{-s/J} \prod_{j=1}^{J} D_{M_j n}^{s/J} \cdot \rho_{n,i} \in \mathbb{Z}, \quad i = 0, 3, 5, \ldots, s-1.$$

By Lemma 5.1, Equation (5.18), and Lemma 5.3, we have

$$\widehat{S}_n = \exp\left(-\widehat{\alpha}(s)n + o(n)\right) \quad \text{as } n \to \infty,$$

where

$$\widehat{\alpha}(s) = \alpha(s) + \frac{s}{J}\varpi - \frac{s}{J}\sum_{j=1}^{J} \max\left\{M-2\delta_1,\ M-\delta_j\right\}.$$

By Lemma 5.2, Equation (5.18), and Lemma 5.3, we have

$$\max_{i=0,3,5,\ldots,s-1} |\widehat{\rho}_{n,i}| \leqslant \exp\left(\widehat{\beta}(s)n + o(n)\right) \quad \text{as } n \to \infty,$$

where

$$\widehat{\beta}(s) = \beta(s) - \frac{s}{J}\varpi + \frac{s}{J}\sum_{j=1}^{J} \max\left\{M-2\delta_1,\ M-\delta_j\right\}.$$

By (5.2) and (5.12), we have the following asymptotic estimates for $\widehat{\alpha}(s)$ and $\widehat{\beta}(s)$ as $s \to \infty$:

$$\widehat{\alpha}(s) \sim \frac{s \log s}{J} \sum_{j=1}^{J} (M - 2\delta_j), \tag{6.2}$$

$$\widehat{\beta}(s) \sim \log 2 \cdot \frac{s}{J} \sum_{j=1}^{J} (M - 2\delta_j) - \frac{s}{J}\varpi + \frac{s}{J} \sum_{j=1}^{J} \max\left\{M - 2\delta_1,\ M - \delta_j\right\}. \tag{6.3}$$

Applying Nesterenko's linear independence criterion (Theorem 2.1) to the sequence of linear forms $\{\widehat{S}_n\}_{n>s^2}$, and using Equations (6.2) and (6.3), we obtain

$$\begin{aligned} \dim_{\mathbb{Q}} \operatorname{Span}_{\mathbb{Q}} (1, \zeta(3), \zeta(5), \ldots, \zeta(s-1)) &\geqslant 1 + \frac{\widehat{\alpha}(s)}{\widehat{\beta}(s)} \\ &= (C - o(1)) \cdot \log s \quad \text{as } s \to \infty, \end{aligned} \tag{6.4}$$

where the constant C is given by

$$C = \frac{\sum_{j=1}^{J} (M - 2\delta_j)}{\log 2 \cdot \sum_{j=1}^{J} (M - 2\delta_j) - \varpi + \sum_{j=1}^{J} \max\{M - 2\delta_1,\ M - \delta_j\}}.$$

We have restricted s to $2J\mathbb{Z}_{>0}$ in our constructions of rational functions and linear forms. Therefore, (6.4) holds as $s \in 2J\mathbb{Z}_{>0}$ and $s \to \infty$. However, if we replace the condition '$s \in 2J\mathbb{Z}_{>0}$' by '$s \in 2\mathbb{Z}_{>0}$', inequality (6.4) remains valid. Thus, the proof of Proposition 6.1 is complete. □

If we take $M = 1$, $\delta_1 = 0$ (with $J = 1$) in Proposition 6.1, then we find $\phi(x) \equiv 0$, $\varpi = 0$, and $C = 1/(1 + \log 2)$. Thus, we recover the Ball–Rivoal theorem [2]. A simple choice of parameters to improve the Ball–Rivoal theorem is: $M = 6$, $\delta_1 = 0$, $\delta_2 = 1$ (with $J = 2$). This collection of parameters leads to the proof of Theorem 1.3:

Proof of Theorem 1.3. Taking $M = 6$, $\delta_1 = 0$, $\delta_2 = 1$ (with $J = 2$) in Proposition 6.1, the function $\phi(\cdot)$ defined by (4.6) becomes

$$\phi(x) = \inf_{y \in \mathbb{R}} \left(\lfloor 6x \rfloor - \lfloor y \rfloor - \lfloor 6x - y \rfloor + \lfloor 4x \rfloor - \lfloor y - x \rfloor - \lfloor 5x - y \rfloor\right).$$

Note that the two-variable function

$$\phi(x, y) := \lfloor 6x \rfloor - \lfloor y \rfloor - \lfloor 6x - y \rfloor + \lfloor 4x \rfloor - \lfloor y - x \rfloor - \lfloor 5x - y \rfloor$$

satisfies $\phi(x, y) = \phi(\{x\}, \{y\})$. In the xy-plane, the lines of the form $6x = k$, $y = k$, $6x - y = k$, $4x = k$, $y - x = k$, and $5x - y = k$ (with $k = 0, \pm 1, \pm 2, \ldots$) divide the unit square $[0, 1]^2$ into several polygons. The function $\phi(x, y)$ is constant in the interior of each polygon. Figure 6.1 illustrates the constant value of $\phi(x, y)$ in the interior of each polygon.

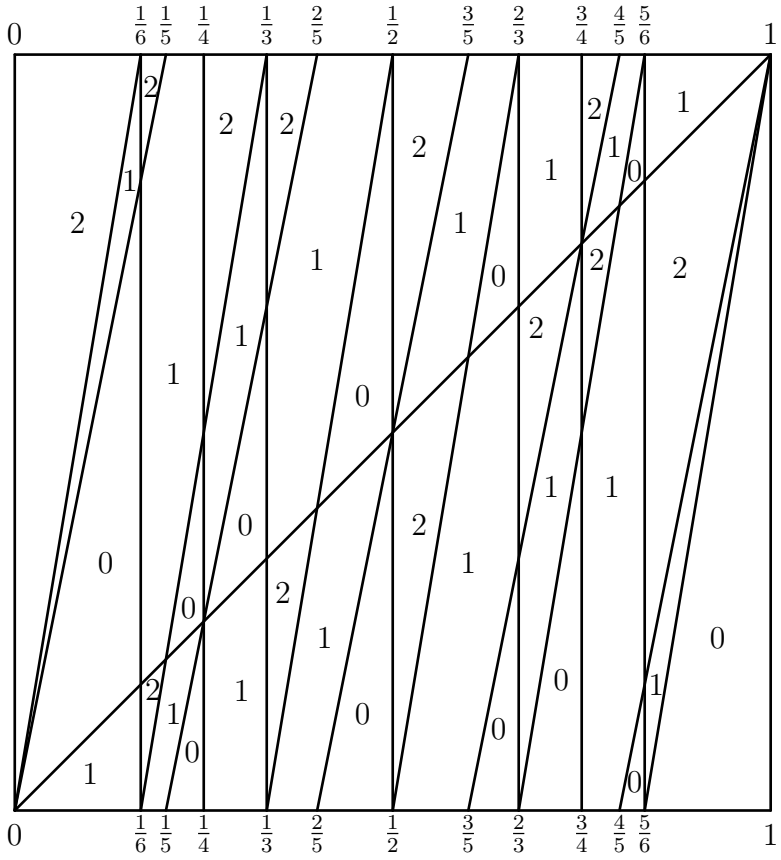


Figure 6.1: The values of $\phi(x, y)$.

Based on Figure 6.1, we can express $\phi(x)$ explicitly as follows:

$$\begin{aligned}\phi(x) &= \inf_{y\in\mathbb{R}} \phi(x,y) = \min_{0\leqslant y<1} \phi(\{x\}, y) \\ &= \begin{cases} 1 & \text{if } \{x\} \in \left[\frac{1}{6}, \frac{1}{5}\right) \cup \left[\frac{1}{3}, \frac{2}{5}\right) \cup \left[\frac{1}{2}, \frac{3}{5}\right) \cup \left[\frac{3}{4}, \frac{4}{5}\right), \\ 0 & \text{otherwise.} \end{cases}\end{aligned}$$

Therefore, by (5.19), we have

$$\begin{aligned}\varpi &= \psi\left(\frac{1}{5}\right) - \psi\left(\frac{1}{6}\right) + \psi\left(\frac{2}{5}\right) - \psi\left(\frac{1}{3}\right) \\ &\quad + \psi\left(\frac{3}{5}\right) - \psi\left(\frac{1}{2}\right) + \psi\left(\frac{4}{5}\right) - \psi\left(\frac{3}{4}\right) \\ &= 2.157479\ldots.\end{aligned}$$

Hence, by (6.1),

$$C = \frac{10}{10\log 2 - \varpi + 12} = \frac{1.009388\ldots}{1+\log 2}.$$

In conclusion, by Proposition 6.1, we have

$$\dim_{\mathbb{Q}} \mathrm{Span}_{\mathbb{Q}} \left(1, \zeta(3), \zeta(5), \ldots, \zeta(s-1)\right) \geqslant \frac{1.009}{1+\log 2} \cdot \log s$$

for any sufficiently large even integer s. The proof of Theorem 1.3 is complete. □

7 Preliminaries for p-adic zeta values

From now on, we fix a prime number p. In this section, we will recall basic facts about Volkenborn integrals, the first Bernoulli functional, p-adic Hurwitz zeta functions, and p-adic zeta values.

7.1 Overconvergent power series on $\mathbb{Z}_p$

Let $C(\mathbb{Z}_p, \mathbb{Q}_p)$ denote the set of all continuous functions $f : \mathbb{Z}_p \longrightarrow \mathbb{Q}_p$. We define the set

$$C^\dagger(\mathbb{Z}_p, \mathbb{Q}_p) := \left\{ f \in C(\mathbb{Z}_p, \mathbb{Q}_p) \;\middle|\; f(t) = \sum_{k=0}^{\infty} a_k t^k, a_k \in \mathbb{Q}_p, \limsup_{k\to\infty} |a_k|_p^{1/k} < 1 \right\}.$$

In other words, $f \in C^\dagger(\mathbb{Z}_p, \mathbb{Q}_p)$ if and only if the function $f(t)$ coincides on $\mathbb{Z}_p$ with a power series in $\mathbb{Q}_p[\![t]\!]$ whose radius of convergence is strictly greater than 1. If $f \in C^\dagger(\mathbb{Z}_p, \mathbb{Q}_p)$, we say that f is an *overconvergent power series*.

If D is a set containing $\mathbb{Z}_p$ and $g : D \longrightarrow \mathbb{Q}_p$ is a function such that the restriction $g|_{\mathbb{Z}_p} \in C^\dagger(\mathbb{Z}_p, \mathbb{Q}_p)$, then by a slight abuse of notation, we write $g \in C^\dagger(\mathbb{Z}_p, \mathbb{Q}_p)$.

7.2 Volkenborn integrals

A function $f : \mathbb{Z}_p \longrightarrow \mathbb{C}_p$ is said to be *Volkenborn integrable* if the following limit exists in $\mathbb{C}_p$:

$$\lim_{N\to\infty} \frac{1}{p^N} \sum_{k=0}^{p^N-1} f(k).$$

For a Volkenborn integrable function f, we define its *Volkenborn integral* by

$$\int_{\mathbb{Z}_p} f(t)\,\mathrm{d}t := \lim_{N\to\infty} \frac{1}{p^N} \sum_{k=0}^{p^N-1} f(k).$$

As usual, we define the Bernoulli numbers $B_k \in \mathbb{Q}$ $(k = 0, 1, 2, \ldots)$ by the generating series

$$\frac{t}{e^t - 1} =: \sum_{k=0}^{\infty} \frac{B_k}{k!} t^k.$$

Lemma 7.1. *Any overconvergent power series $f(t) = \sum_{k=0}^{\infty} a_k t^k \in C^\dagger(\mathbb{Z}_p, \mathbb{Q}_p)$ is Volkenborn integrable. Moreover, we have*

$$\int_{\mathbb{Z}_p} f(t)\,\mathrm{d}t = \sum_{k=0}^{\infty} a_k B_k.$$

Proof. See [13, Proposition, p. 270], where a stronger result is proved. □

The Volkenborn integral has the following behavior under translations:

Lemma 7.2 ([13, Proposition 2, p. 265])**.** *Let $k \in \mathbb{Z}_{>0}$ and $f \in C^\dagger(\mathbb{Z}_p, \mathbb{Q}_p)$. Then, we have*

$$\int_{\mathbb{Z}_p} f(t+k)\,\mathrm{d}t = \int_{\mathbb{Z}_p} f(t)\,\mathrm{d}t + \sum_{\ell=0}^{k-1} f'(\ell).$$

7.3 The first Bernoulli functional $\mathcal{L}_1$

The following definition turns out to be useful:

Definition 7.3 ([8, Definition 2.6])**.** We define the *first Bernoulli functional*

$$\mathcal{L}_1 : C^\dagger(\mathbb{Z}_p, \mathbb{Q}_p) \longrightarrow \mathbb{Q}_p$$

by

$$f = \sum_{k=0}^{\infty} a_k t^k \mapsto \mathcal{L}_1(f) := \sum_{k=0}^{\infty} \frac{a_k B_{k+1}}{k+1}.$$

Since $\limsup_{k\to\infty} |a_k|_p^{1/k} < 1$ and $|B_{k+1}/(k+1)|_p \leqslant p(k+1)$ (by the von Staudt–Clausen theorem), the sum $\sum_{k=0}^{\infty} a_k B_{k+1}/(k+1)$ is convergent in $\mathbb{Q}_p$. The main property of $\mathcal{L}_1$ is the following:

Lemma 7.4 ([8, Lemma 2.7])**.** *For any $f \in C^\dagger(\mathbb{Z}_p, \mathbb{Q}_p)$ we have the formula*

$$\mathcal{L}_1(f') = \int_{\mathbb{Z}_p} f(t)\,\mathrm{d}t - f(0).$$

Proof. We have $f' \in C^\dagger(\mathbb{Z}_p, \mathbb{Q}_p)$ because $C^\dagger(\mathbb{Z}_p, \mathbb{Q}_p)$ is clearly closed under differentiation. Suppose that $f(t) = \sum_{k=0}^{\infty} a_k t^k$. Then, by Definition 7.3 and Lemma 7.1, we have

$$\mathcal{L}_1(f') = \sum_{k=1}^{\infty} a_k B_k = \int_{\mathbb{Z}_p} f(t)\,\mathrm{d}t - f(0).$$ □

Remark 7.5. The first Bernoulli functional $\mathcal{L}_1$ can be viewed as a de-regularization of the first Bernoulli measure $\mu_{1,\alpha}$ over the space of overconvergent power series, as noted in [8, Proposition 2.9]. This justifies the name of $\mathcal{L}_1$.

7.4 p-Adic Hurwitz zeta functions and p-adic zeta values

In this subsection, we will recall some basic facts about p-adic Hurwitz zeta functions and p-adic zeta values.

Set $q_p = p$ if $p > 2$ and $q_2 = 4$. The units $\mathbb{Z}_p^\times$ of the p-adic integers decompose canonically as follows:

$$\mathbb{Z}_p^\times \cong \mu_{\varphi(q_p)}(\mathbb{Z}_p) \times (1 + q_p\mathbb{Z}_p).$$

Here, $\mu_{\varphi(q_p)}(\mathbb{Z}_p)$ denotes the group of $\varphi(q_p)$-th roots of unity in $\mathbb{Z}_p$. The canonical projection

$$\omega\colon \mathbb{Z}_p^\times \longrightarrow \mu_{\varphi(q_p)}(\mathbb{Z}_p)$$

is called the *Teichmüller character*, which can be extended multiplicatively to a map

$$\mathbb{Q}_p^\times \longrightarrow \mathbb{Q}_p^\times$$

by setting

$$\omega(x) := p^{v_p(x)}\omega\left(\frac{x}{p^{v_p(x)}}\right).$$

We define

$$\langle x\rangle := \frac{x}{\omega(x)}$$

for $x \in \mathbb{Q}_p^\times$. Both $\omega(x)$ and $\langle x\rangle$ are multiplicative on $\mathbb{Q}_p^\times$. Moreover, we have $\langle x\rangle \in 1+q_p\mathbb{Z}_p$ for any $x \in \mathbb{Q}_p^\times$.

For any $x \in \mathbb{Q}_p$ with $|x|_p \geqslant q_p$, and for any $s \in \mathbb{C}_p \setminus \{1\}$ with $|s|_p < q_p p^{-1/(p-1)}$, we define

$$\zeta_p(s,x) := \frac{1}{s-1}\int_{\mathbb{Z}_p} \langle t+x\rangle^{1-s}\,\mathrm{d}t. \tag{7.1}$$

The function

$$\zeta_p(\cdot,x)\colon \left\{s \in \mathbb{C}_p \setminus \{1\} \;\middle|\; |s|_p < q_p p^{-1/(p-1)}\right\} \longrightarrow \mathbb{C}_p$$

is called the *p-adic Hurwitz zeta function*. It is known that $\zeta_p(\cdot,x)$ is a p-adic meromorphic function on the region $|s|_p < q_p p^{-1/(p-1)}$, as stated in [4, Proposition 11.2.8].

Let χ be a Dirichlet character of conductor D. Let m be a common multiple of D and q_p. The *Kubota–Leopoldt p-adic L-function* is defined by

$$L_p(s,\chi) := \frac{\langle m\rangle^{1-s}}{m}\sum_{\substack{1\leqslant\nu\leqslant m\\ p\nmid\nu}} \chi(\nu)\zeta_p\left(s,\frac{\nu}{m}\right) \tag{7.2}$$

for $s \in \mathbb{C}_p \setminus \{1\}$ with $|s|_p < q_p p^{-1/(p-1)}$. It is known that the right-hand side of (7.2) is independent of m (see [4, Proposition 11.3.8(1)]).

Definition 7.6. For any integer $s \geqslant 2$, we define the *p-adic zeta value* $\zeta_p(s)$ by

$$\zeta_p(s) := L_p(s,\omega^{1-s}).$$

We remark that

$$\zeta_p(s) = \lim_{\substack{k \to s \ p\text{-adically} \\ k \in \mathbb{Z}_{<0}, \quad k \equiv s \pmod{p-1}}} \zeta(k) \quad \in \mathbb{Q}_p$$

for any integer $s \geqslant 2$ (see [3, Lemma 2.4]). Moreover, $\zeta_p(s)$ vanishes when s is a positive even integer.

For our purposes, we need the following lemmas.

Lemma 7.7. *Let i, l be integers such that $i \geqslant 2$ and $l \geqslant 2$. Let ν be an integer such that $p \nmid \nu$. Let k be any nonnegative integer. Then, we have*

$$-\int_{\mathbb{Z}_p} \frac{1}{(1-i)(t+\nu/p^l+k)^{i-1}}\,\mathrm{d}t = \omega\left(\frac{\nu}{p^l}\right)^{1-i} \zeta_p\left(i, \frac{\nu}{p^l}\right) - \sum_{\ell=0}^{k-1} \frac{1}{(\ell+\nu/p^l)^i}.$$

(*Note: When $k = 0$, the empty sum $\sum_{\ell=0}^{k-1}$ is understood as* 0.)

Proof. By Lemma 7.2 and Equation (7.1), we have

$$\begin{aligned}
&-\int_{\mathbb{Z}_p} \frac{1}{(1-i)(t+\nu/p^l+k)^{i-1}}\,\mathrm{d}t \\
&= -\int_{\mathbb{Z}_p} \frac{1}{(1-i)(t+\nu/p^l)^{i-1}}\,\mathrm{d}t - \sum_{\ell=0}^{k-1} \frac{1}{(\ell+\nu/p^l)^i} \\
&= \omega\left(\frac{\nu}{p^l}\right)^{1-i} \zeta_p\left(i, \frac{\nu}{p^l}\right) - \sum_{\ell=0}^{k-1} \frac{1}{(\ell+\nu/p^l)^i}.
\end{aligned}$$

□

Lemma 7.8. *Let i, l be integers such that $i \geqslant 2$ and $l \geqslant 2$. Then, we have*

$$p^{li} \cdot \zeta_p(i) = \sum_{\substack{1 \leqslant \nu \leqslant p^l \\ p \nmid \nu}} \omega\left(\frac{\nu}{p^l}\right)^{1-i} \zeta_p\left(i, \frac{\nu}{p^l}\right).$$

Proof. It follows from Definition 7.6 and Equation (7.2) (with $m = p^l$). □

8 Setup for the proof for the p-adic case

In this section, we will first set up the parameters. Then, we will construct a sequence of rational functions $V_n(t)$ and utilize their primitive functions $W_n(t)$ to produce linear forms in 1 and p-adic odd zeta values.

8.1 Parameters

Fix a positive integer M, and fix a finite collection of nonnegative integers $\{\delta_j\}_{j=1}^{J}$ such that

$$0 \leqslant \delta_1 \leqslant \delta_2 \leqslant \cdots \leqslant \delta_J < \frac{M}{2}, \tag{8.1}$$

where $J \geqslant 1$ denotes the number of δ_j's.

Let s be a sufficiently large multiple of $2J$. Define the 'power parameter' l as

$$l := \left\lfloor \frac{\log s - 2\log\log s}{\log p} \right\rfloor. \tag{8.2}$$

We assume that s is sufficiently large so that $l \geqslant 2$. Note that l is defined in such a way that p^l is the largest power of p not exceeding $s/\log^2 s$. The role of the parameter l in the p-adic case is similar to that of the 'numerator length parameter' r in the classical case.

8.2 Rational functions and their primitive functions

For any $n \in \mathbb{Z}_{>0}$, we define

$$L_n := \left\lfloor \frac{\log(M\varphi(p^l)n)}{\log p} \right\rfloor + 2. \tag{8.3}$$

Let w_n be the unique integer satisfying

$$\begin{aligned} &w_n \equiv s - 1 + \left(-M\varphi(p^l) + \frac{s}{J}\sum_{j=1}^{J}(M - 2\delta_j) \right) n \pmod{(p-1)p^{L_n}}, \\ &0 \leqslant w_n < (p-1)p^{L_n}. \end{aligned} \tag{8.4}$$

Note that

$$0 \leqslant w_n < p^{l+3}Mn. \tag{8.5}$$

The definitions of L_n and w_n are technical. They are used in Section 11 to prove a certain nonvanishing property (see Lemma 11.2).

For any $n \in \mathbb{Z}_{>0}$, we define the following rational function $V_n(t)$:

$$\begin{aligned} V_n(t) := p^{(l+1/(p-1))M\varphi(p^l)n} \cdot \frac{\prod_{j=1}^{J}((M-2\delta_j)n)!^{s/J}}{n!^{M\varphi(p^l)}} \cdot \frac{1}{n!^{\lfloor w_n/n \rfloor}(w_n - \lfloor w_n/n \rfloor n)!} \\ \times (t - w_n)_{w_n} \cdot \frac{\prod_{\substack{1 \leqslant \nu \leqslant p^l \\ p \nmid \nu}} (t + \nu/p^l)_{Mn}}{\prod_{j=1}^{J}(t+\delta_j n)_{(M-2\delta_j)n+1}^{s/J}}. \end{aligned} \tag{8.6}$$

Since $J \mid s$ and $(p-1) \mid \varphi(p^l)$, we have $V_n(t) \in \mathbb{Q}(t)$. Note that the degree of the rational function $V_n(t)$ is

$$\deg V_n = w_n + M\varphi(p^l)n - s - \frac{s}{J}\sum_{j=1}^{J}(M - 2\delta_j)n \tag{8.7}$$

$$< p^{l+3}Mn + M\varphi(p^l)n - s - sn \quad \text{(by (8.5) and (8.1))}$$
$$< \frac{p^3 Msn}{\log^2 s} + \frac{Msn}{\log^2 s} - s - sn. \quad \text{(by (8.2))}$$

We assume that s is sufficiently large so that

$$\deg V_n \leqslant -2 \quad \text{for any } n \in \mathbb{Z}_{>0}. \tag{8.8}$$

Therefore, the rational function $V_n(t)$ has the following partial-fraction decomposition:

$$V_n(t) =: \sum_{i=1}^{s} \sum_{k=\delta_1 n}^{(M-\delta_1)n} \frac{b_{n,i,k}}{(t+k)^i}, \tag{8.9}$$

where the coefficients $b_{n,i,k} \in \mathbb{Q}$ ($1 \leqslant i \leqslant s$ and $\delta_1 n \leqslant k \leqslant (M-\delta_1)n$) are uniquely determined by $V_n(t)$.

For any $n \in \mathbb{Z}_{>0}$, we define the function $W_n(t)$ by

$$W_n(t) := \sum_{k=\delta_1 n}^{(M-\delta_1)n} b_{n,1,k} \log_p \langle t+k \rangle + \sum_{i=2}^{s} \sum_{k=\delta_1 n}^{(M-\delta_1)n} \frac{b_{n,i,k}}{(1-i)(t+k)^{i-1}}. \tag{8.10}$$

The function $W_n(t)$ has has the following properties, which motivate its definition.

Lemma 8.1. *For any $n \in \mathbb{Z}_{>0}$, the function $W_n(t)$ has the following properties.*

(1) *$W_n(t)$ is a primitive function of $V_n(t)$ on $\mathbb{Q}_p \setminus ([-(M-\delta_1)n, -\delta_1 n] \cap \mathbb{Z})$; that is,*

$$W_n'(t) = V_n(t), \quad t \in \mathbb{Q}_p \setminus ([-(M-\delta_1)n, -\delta_1 n] \cap \mathbb{Z}).$$

(2) *For any integer $\nu \in [1, p^l]$ with $p \nmid \nu$, the restriction of $W_n(t + \nu/p^l)$ to $\mathbb{Z}_p$ is an overconvergent power series; that is,*

$$W_n\left(t + \frac{\nu}{p^l}\right) \in C^\dagger(\mathbb{Z}_p, \mathbb{Q}_p).$$

(3) *For $|t|_p \geqslant 1$, the function $W_n(t/p^l)$ can be expressed as a $\mathbb{Q}_p$-coefficient convergent power series in t^{-1} with vanishing constant term.*

Proof. (1). By (8.10) and (8.9), we have

$$W_n'(t) = \sum_{k=\delta_1 n}^{(M-\delta_1)n} \frac{b_{n,1,k}}{t+k} + \sum_{i=2}^{s} \sum_{k=\delta_1 n}^{(M-\delta_1)n} \frac{b_{n,i,k}}{(t+k)^i} = V_n(t)$$

for $t \in \mathbb{Q}_p \setminus ([-(M-\delta_1)n, -\delta_1 n] \cap \mathbb{Z})$.

(2). For $t \in \mathbb{Z}_p$, we have

$$\left\langle t + \frac{\nu}{p^l} + k \right\rangle = \left\langle \frac{\nu}{p^l} + k \right\rangle \langle 1 + (\nu + p^l k)^{-1} p^l t \rangle$$

$$= \left\langle \frac{\nu}{p^l} + k \right\rangle (1 + (\nu + p^l k)^{-1} p^l t).$$

Thus, by (8.10), we have

$$\begin{aligned} & W_n\left(t + \frac{\nu}{p^l}\right) \\ = & \underbrace{\sum_{k=\delta_1 n}^{(M-\delta_1)n} b_{n,1,k} \log_p \left\langle \frac{\nu}{p^l} + k \right\rangle}_{\text{constant}} + \sum_{k=\delta_1 n}^{(M-\delta_1)n} b_{n,1,k} \underbrace{\log_p(1 + (\nu + p^l k)^{-1} p^l t)}_{\in C^\dagger(\mathbb{Z}_p, \mathbb{Q}_p)} \\ & + \sum_{i=2}^{s} \sum_{k=\delta_1 n}^{(M-\delta_1)n} \frac{b_{n,i,k}}{(1-i)(\nu/p^l + k)^{i-1}} \underbrace{(1 + (\nu + p^l k)^{-1} p^l t)^{1-i}}_{\in C^\dagger(\mathbb{Z}_p, \mathbb{Q}_p)}. \end{aligned}$$

Every summand on the right-hand side is an overconvergent power series. Therefore, $W_n(t + \nu/p^l) \in C^\dagger(\mathbb{Z}_p, \mathbb{Q}_p)$.

(3). For $|t|_p \geqslant 1$, we have

$$\left\langle \frac{t}{p^l} + k \right\rangle = \langle t + p^l k \rangle = \langle t \rangle (1 + p^l k t^{-1}).$$

Thus, by (8.10), we have

$$\begin{aligned} W_n\left(\frac{t}{p^l}\right) = & \left(\sum_{k=\delta_1 n}^{(M-\delta_1)n} b_{n,1,k} \right) \log_p \langle t \rangle + \sum_{k=\delta_1 n}^{(M-\delta_1)n} b_{n,1,k} \log_p(1 + p^l k t^{-1}) \\ & + \sum_{i=2}^{s} \sum_{k=\delta_1 n}^{(M-\delta_1)n} \frac{p^{l(i-1)} b_{n,i,k}}{(1-i)} t^{1-i} (1 + p^l k t^{-1})^{1-i}. \end{aligned}$$

By (8.9) and (8.8), we have

$$\sum_{k=\delta_1 n}^{(M-\delta_1)n} b_{n,1,k} = \lim_{|t|_p \to +\infty} t V_n(t) = 0. \tag{8.11}$$

Therefore, for $|t|_p \geqslant 1$,

$$\begin{aligned} W_n\left(\frac{t}{p^l}\right) = & \sum_{k=\delta_1 n}^{(M-\delta_1)n} b_{n,1,k} \underbrace{\log_p(1 + p^l k t^{-1})}_{\in \mathbb{Q}_p[\![p^l t^{-1}]\!]} \\ & + \sum_{i=2}^{s} \sum_{k=\delta_1 n}^{(M-\delta_1)n} \frac{p^{l(i-1)} b_{n,i,k}}{(1-i)} t^{1-i} \underbrace{(1 + p^l k t^{-1})^{1-i}}_{\in \mathbb{Q}_p[\![p^l t^{-1}]\!]} \end{aligned}$$

is a $\mathbb{Q}_p$-coefficient convergent power series in t^{-1} with vanishing constant term. □

8.3 Linear forms

By Lemma 8.1(2), for any integer $\nu \in [1, p^l]$ with $p \nmid \nu$, we have $W_n\left(t + \nu/p^l\right) \in C^\dagger(\mathbb{Z}_p, \mathbb{Q}_p)$. Therefore, $W_n\left(t + \nu/p^l\right)$ is Volkenborn integrable by Lemma 7.1.

Definition 8.2. For any $n \in \mathbb{Z}_{>0}$, for any integer $\nu \in [1, p^l]$ with $p \nmid \nu$, we define

$$T_{n,\nu/p^l} := -\int_{\mathbb{Z}_p} W_n\left(t + \frac{\nu}{p^l}\right) \mathrm{d}t. \tag{8.12}$$

Moreover, we define

$$T_n := \sum_{\substack{1 \leqslant \nu \leqslant p^l \\ p \nmid \nu}} T_{n,\nu/p^l}. \tag{8.13}$$

We define the following coefficients:

$$\sigma_{n,i} := \sum_{k=\delta_1 n}^{(M-\delta_1)n} b_{n,i,k}, \quad i = 1, 2, 3, \ldots, s, \tag{8.14}$$

$$\sigma_{n,0,\nu/p^l} := -\sum_{i=1}^{s} \sum_{k=\delta_1 n}^{(M-\delta_1)n} \sum_{\ell=0}^{k-1} \frac{b_{n,i,k}}{(\ell + \nu/p^l)^i}, \quad 1 \leqslant \nu \leqslant p^l \text{ with } p \nmid \nu. \tag{8.15}$$

$$\sigma_{n,0} := \sum_{\substack{1 \leqslant \nu \leqslant p^l \\ p \nmid \nu}} \sigma_{n,0,\nu/p^l}. \tag{8.16}$$

(Note: When $k = 0$, the empty sum $\sum_{\ell=0}^{k-1}$ in (8.15) is understood to be 0.)

It turns out that T_n is a linear form in 1 and p-adic odd zeta values.

Lemma 8.3. *For any $n \in \mathbb{Z}_{>0}$, T_n is a linear form in 1 and p-adic odd zeta values, given by*

$$T_n = \sigma_{n,0} + \sum_{\substack{3 \leqslant i \leqslant s-1 \\ i \text{ odd}}} \sigma_{n,i} p^{li} \zeta_p(i),$$

where the coefficients $\sigma_{n,i}$ $(i = 3, 5, \ldots, s-1)$ are defined by (8.14) *and $\sigma_{n,0}$ is defined by* (8.16).

Proof. For any integer $\nu \in [1, p^l]$ with $p \nmid \nu$, by (8.12) and (8.10), we have

$$\begin{aligned} T_{n,\nu/p^l} = &-\sum_{k=\delta_1 n}^{(M-\delta_1)n} b_{n,1,k} \int_{\mathbb{Z}_p} \log_p\left\langle t + \frac{\nu}{p^l} + k \right\rangle \mathrm{d}t \\ &-\sum_{i=2}^{s} \sum_{k=\delta_1 n}^{(M-\delta_1)n} b_{n,i,k} \int_{\mathbb{Z}_p} \frac{1}{(1-i)(t + \nu/p^l + k)^{i-1}} \, \mathrm{d}t. \end{aligned} \tag{8.17}$$

By Lemma 7.7, we have

$$-\int_{\mathbb{Z}_p} \frac{1}{(1-i)(t + \nu/p^l + k)^{i-1}} \, \mathrm{d}t = \omega\left(\frac{\nu}{p^l}\right)^{1-i} \zeta_p\left(i, \frac{\nu}{p^l}\right) - \sum_{\ell=0}^{k-1} \frac{1}{(\ell + \nu/p^l)^i}. \tag{8.18}$$

By Lemma 7.2, we have

$$-\int_{\mathbb{Z}_p} \log_p \left\langle t+\frac{\nu}{p^l}+k \right\rangle \mathrm{d}t = -\int_{\mathbb{Z}_p} \log_p \left\langle t+\frac{\nu}{p^l} \right\rangle \mathrm{d}t - \sum_{\ell=0}^{k-1} \frac{1}{\ell+\nu/p^l}. \tag{8.19}$$

Substituting (8.18) and (8.19) into (8.17), we obtain

$$\begin{aligned} T_{n,\nu/p^l} = &-\sigma_{n,1} \int_{\mathbb{Z}_p} \log_p \left\langle t+\frac{\nu}{p^l} \right\rangle \mathrm{d}t \\ &+ \sigma_{n,0,\nu/p^l} + \sum_{i=2}^{s} \sigma_{n,i}\omega\left(\frac{\nu}{p^l}\right)^{1-i} \zeta_p\left(i, \frac{\nu}{p^l}\right). \end{aligned}$$

Since $\sigma_{n,1}=0$ by (8.11), we have

$$T_{n,\nu/p^l} = \sigma_{n,0,\nu/p^l} + \sum_{i=2}^{s} \sigma_{n,i}\omega\left(\frac{\nu}{p^l}\right)^{1-i} \zeta_p\left(i, \frac{\nu}{p^l}\right). \tag{8.20}$$

Finally, substituting (8.20) into (8.13) and using Lemma 7.8, we obtain

$$T_n = \sigma_{n,0} + \sum_{i=2}^{s} \sigma_{n,i} p^{li} \zeta_p(i).$$

Since $\zeta_p(i)=0$ for any positive even integer i, this completes the proof of Lemma 8.3. □

9 Arithmetic properties for the p-adic case

Recall that $D_m = \mathrm{lcm}[1,2,3,\ldots,m]$. The following lemma is analogous to Lemma 4.3:

Lemma 9.1. *For any integer $n > s^2$, we have*

$$\Phi_n^{-s/J} D_{(M-2\delta_1)n}^{s-i} \cdot b_{n,i,k} \in \mathbb{Z} \quad (1 \leqslant i \leqslant s \text{ and } \delta_1 n \leqslant k \leqslant (M-\delta_1)n),$$

where the factor Φ_n is given by (4.5) *and* (4.6).

Proof. The proof is similar to that of Lemma 4.3. We indicate the modifications.

Fix any $i \in \{1,2,\ldots,s\}$ and any $k \in [\delta_1 n,\ (M-\delta_1)n] \cap \mathbb{Z}$.

By (8.9), we have

$$b_{n,i,k} = \frac{1}{(s-i)!} \left(V_n(t)(t+k)^s\right)^{(s-i)} \Big|_{t=-k}.$$

Define the elementary bricks:

$$\begin{aligned} F_{1,\tau}(t) &= \frac{(t-w_n+\tau n)_n}{n!}, \quad \tau = 0,1,2,\ldots,\lfloor w_n/n \rfloor - 1, \\ F_{1,\lfloor w_n/n \rfloor}(t) &= \frac{(t-w_n+\lfloor w_n/n \rfloor n)_{w_n-\lfloor w_n/n \rfloor n}}{(w_n-\lfloor w_n/n \rfloor n)!}, \end{aligned}$$

$$F_{\nu/p^l,\tau}(t) = p^{ln} \cdot p^{\lfloor n/(p-1)\rfloor} \cdot \frac{(t+\nu/p^l+\tau n)_n}{n!},$$
$$1 \leqslant \nu \leqslant p^l \text{ with } p \nmid \nu, \quad \tau = 0, 1, 2, \ldots, M-1,$$
$$G_j(t) = \frac{((M-2\delta_j)n)!}{(t+\delta_j n)_{(M-2\delta_j)n+1}}, \quad j = 1, 2, \ldots, J.$$

Then, by (8.6), we have

$$V_n(t)(t+k)^s = A \prod_{\tau=0}^{\lfloor w_n/n\rfloor} F_{1,\tau}(t) \cdot \prod_{\substack{1\leqslant\nu\leqslant p^l\\ p\nmid\nu}} \prod_{\tau=0}^{M-1} F_{\nu/p^l,\tau}(t) \cdot \prod_{j=1}^{J} \left(G_j(t)(t+k)\right)^{s/J},$$

where $A = p^{M\varphi(p^l)\{n/(p-1)\}}$ is a positive integer. By Lemma 4.2, for any polynomial $F(t)$ of the form $F_{1,\tau}(t)$, $F_{\nu/p^l,\tau}(t)$, we have

$$D_n^\lambda \cdot \frac{1}{\lambda!} F^{(\lambda)}(t)\big|_{t=-k} \in \mathbb{Z}$$

for any nonnegative integer λ. By (4.1) of Lemma 4.1 (with $a_0 = \delta_1 n$ and $b_0 = (M-\delta_1)n$), we have

$$D_{(M-2\delta_1)n}^\lambda \cdot \frac{1}{\lambda!} \left(G_j(t)(t+k)\right)^{(\lambda)}\big|_{t=-k} \in \mathbb{Z}$$

for any $j = 1, 2, \ldots, J$ and any nonnegative integer λ.

The rest of the proof of Lemma 9.1 is literally the same as the rest of the proof of Lemma 4.3, except that we need to replace $a_{n,i,k}$ by $b_{n,i,k}$. □

The arithmetic property of $b_{n,i,k}$ propagates to the arithmetic property of $\sigma_{n,i}$ (defined by (8.14) and (8.16)), as the following lemma shows.

Lemma 9.2. *For any integer $n > s^2$, we have*

$$\Phi_n^{-s/J} D_{(M-2\delta_1)n}^{s-i} \cdot \sigma_{n,i} \in \mathbb{Z} \quad (i = 1, 2, \ldots, s),$$

and

$$\Phi_n^{-s/J} D_{(M-2\delta_1)n}^{s} \cdot \sigma_{n,0} \in \mathbb{Z}.$$

Proof. The first assertion follows directly from (8.14) and Lemma 9.1.

To prove the second assertion, by (8.16), it suffices to prove that for any integer $\nu \in [1, p^l]$ with $p \nmid \nu$ we have

$$\Phi_n^{-s/J} D_{(M-2\delta_1)n}^{s} \cdot \sigma_{n,0,\nu/p^l} \in \mathbb{Z}. \tag{9.1}$$

We prove (9.1) by contradiction. Suppose that

$$\Phi_n^{-s/J} D_{(M-2\delta_1)n}^{s} \cdot \sigma_{n,0,\nu/p^l} \notin \mathbb{Z}. \tag{9.2}$$

By substituting (8.15) into (9.2), we see that there exist integers k_0, ℓ_0 such that $\delta_1 n \leqslant k_0 \leqslant (M-\delta_1)n$, $0 \leqslant \ell_0 < k_0$, and

$$\Phi_n^{-s/J} D_{(M-2\delta_1)n}^{s} \sum_{i=1}^{s} \frac{b_{n,i,k_0}}{(\ell_0+\nu/p^l)^i} \notin \mathbb{Z}. \tag{9.3}$$

Write $\nu' = p^l - \nu$. Note that $t + k_0 - \ell_0 - \nu/p^l$ is a factor of $(t + \nu'/p^l)_{Mn}$. Thus, $t + k_0 - \ell_0 - \nu/p^l$ is a factor of the numerator of $V_n(t)$ (see (8.6)). We have

$$V_n\left(-k_0 + \ell_0 + \frac{\nu}{p^l}\right) = 0. \tag{9.4}$$

Substituting (8.9) into (9.4), and using (9.3), we obtain

$$\begin{aligned}
&\Phi_n^{-s/J} D^s_{(M-2\delta_1)n} \sum_{i=1}^{s} \frac{b_{n,i,k_0}}{(\ell_0 + \nu/p^l)^i} \\
= \; &-\Phi_n^{-s/J} D^s_{(M-2\delta_1)n} \sum_{i=1}^{s} \sum_{\substack{k=\delta_1 n \\ k\neq k_0}}^{(M-\delta_1)n} \frac{b_{n,i,k}}{(k - k_0 + \ell_0 + \nu/p^l)^i} \notin \mathbb{Z}.
\end{aligned}$$

Therefore, there exist a prime q, two integers $i_0, i_1 \in \{1, 2, \dots, s\}$, and an integer $k_1 \in \{\delta_1 n, \dots, (M - \delta_1)n\}$ with $k_1 \neq k_0$ such that

$$\begin{aligned}
v_q\left(\Phi_n^{-s/J} D^s_{(M-2\delta_1)n} \cdot \frac{b_{n,i_0,k_0}}{(\ell_0 + \nu/p^l)^{i_0}}\right) < 0 \quad \text{and} \\
v_q\left(\Phi_n^{-s/J} D^s_{(M-2\delta_1)n} \cdot \frac{b_{n,i_1,k_1}}{(k_1 - k_0 + \ell_0 + \nu/p^l)^{i_1}}\right) < 0.
\end{aligned}$$

On the other hand, by Lemma 9.1, we have

$$v_q\left(\Phi_n^{-s/J} D^{s-i_0}_{(M-2\delta_1)n} \cdot b_{n,i_0,k_0}\right) \geqslant 0 \quad \text{and} \quad v_q\left(\Phi_n^{-s/J} D^{s-i_1}_{(M-2\delta_1)n} \cdot b_{n,i_1,k_1}\right) \geqslant 0.$$

Therefore,

$$v_q\left(\ell_0 + \frac{\nu}{p^l}\right) > v_q(D_{(M-2\delta_1)n}) \quad \text{and} \quad v_q\left(k_1 - k_0 + \ell_0 + \frac{\nu}{p^l}\right) > v_q(D_{(M-2\delta_1)n}).$$

It follows that $v_q(k_1 - k_0) > v_q(D_{(M-2\delta_1)n})$, which is a contradiction because $0 < |k_1 - k_0| \leqslant (M - 2\delta_1)n$. The proof of Lemma 9.2 is complete. □

10 Archimedean estimates

In this section we will estimate the Archimedean norm of the coefficients $\sigma_{n,i}$ ($i = 0, 1, 2, \dots, s$) defined by (8.16) and (8.14).

Lemma 10.1. *We have*

$$\max_{0\leqslant i\leqslant s} |\sigma_{n,i}| \leqslant \exp\left(\beta_p(s)n + o(n)\right) \quad \text{as } n \to \infty, \tag{10.1}$$

where $\beta_p(s)$ is a constant independent of n. Moreover, we have

$$\beta_p(s) \sim \log 2 \cdot \frac{s}{J} \sum_{j=1}^{J} (M - 2\delta_j) \quad \text{as } s \to \infty. \tag{10.2}$$

Proof. By (8.14)–(8.16), we have

$$\max_{0\leqslant i\leqslant s}|\sigma_{n,i}|\leqslant (Mn+1)^2 sp^{l(s+1)}\cdot \max_{i,k}|b_{n,i,k}|, \tag{10.3}$$

where $\max_{i,k}$ is taken over all $i\in\{1,2,\dots,s\}$ and $k\in[\delta_1 n,(M-\delta_1)n]\cap\mathbb{Z}$.

Now fix any pair (i,k) in the range above. By (8.9) and Cauchy's integral formula, we have

$$b_{n,i,k}=\frac{1}{2\pi\sqrt{-1}}\int_{|z+k|=1/p^{l+1}}(z+k)^{i-1}V_n(z)\,\mathrm{d}z.$$

Therefore, we have (recall (8.6))

$$\begin{aligned}
|b_{n,i,k}|&\leqslant \max_{|z+k|=1/p^{l+1}}|V_n(z)|\\
&=p^{(l+1/(p-1))M\varphi(p^l)n}\cdot\frac{\prod_{j=1}^J((M-2\delta_j)n)!^{s/J}}{n!^{M\varphi(p^l)}}\cdot\frac{1}{n!^{\lfloor w_n/n\rfloor}(w_n-\lfloor w_n/n\rfloor n)!}\\
&\quad\times\max_{|z+k|=1/p^{l+1}}|(z-w_n)_{w_n}|\frac{\prod_{\substack{1\leqslant\nu\leqslant p^l\\ p\nmid\nu}}|(z+\nu/p^l)_{Mn}|}{\prod_{j=1}^J|(z+\delta_j n)_{(M-2\delta_j)n+1}|^{s/J}}.
\end{aligned}\tag{10.4}$$

In the following, the complex number z is always assumed to be on the circle $|z+k|=1/p^{l+1}$. We will use the triangle inequality to bound (10.4) from above.

We have

$$\begin{aligned}
&\max_{|z+k|=1/p^{l+1}}\frac{|(z-w_n)_{w_n}|}{n!^{\lfloor w_n/n\rfloor}(w_n-\lfloor w_n/n\rfloor n)!}\\
&\leqslant\frac{(Mn+2)(Mn+3)\cdots(Mn+w_n+1)}{n!^{\lfloor w_n/n\rfloor}(w_n-\lfloor w_n/n\rfloor n)!}\\
&=\frac{(Mn+w_n+1)!}{(Mn+1)!n!^{\lfloor w_n/n\rfloor}(w_n-\lfloor w_n/n\rfloor n)!}\\
&\leqslant\left(\underbrace{1+1+\cdots+1}_{\lfloor w_n/n\rfloor+2}\right)^{Mn+w_n+1}\\
&\leqslant(p^{l+3}M+2)^{Mn+p^{l+3}Mn+1}.\quad\text{(by (8.5))}
\end{aligned}\tag{10.5}$$

For any integer $\nu\in[1,p^l]$ with $p\nmid\nu$, we have

$$\begin{aligned}
\max_{|z+k|=1/p^{l+1}}\frac{|(z+\nu/p^l)_{Mn}|}{n!^M}&\leqslant\frac{(k+1)!(Mn+1-k)!}{n!^M}\\
&\leqslant\frac{(Mn+2)!}{n!^M}=(Mn+2)(Mn+1)\cdot\frac{(Mn)!}{n!^M}\\
&\leqslant 10M^2n^2\cdot M^{Mn}.
\end{aligned}\tag{10.6}$$

Arguing as in the proof of (5.17), we obtain

$$\max_{|z+k|=1/p^{l+1}}\frac{((M-2\delta_j)n)!}{|(z+\delta_j n)_{(M-2\delta_j)n+1}|}\leqslant p^{3(l+1)}M^2n^2\cdot 2^{(M-2\delta_j)n}.\tag{10.7}$$

Combining inequalities (10.3)–(10.7), we obtain

$$\max_{0\leqslant i\leqslant s}|\sigma_{n,i}|\leqslant \exp\left(\beta_p(s)n+o(n)\right)\quad \text{as } n\to\infty,$$

where

$$\begin{aligned}\beta_p(s)=&\left(l+\frac{1}{p-1}\right)M\varphi(p^l)\log p+M\varphi(p^l)\log M\\&+(p^{l+3}+1)M\log(p^{l+3}M+2)+\log 2\cdot\frac{s}{J}\sum_{j=1}^{J}(M-2\delta_j).\end{aligned}$$

By (8.2), we have

$$\varphi(p^l)<p^l\leqslant\frac{s}{\log^2 s}\quad\text{and}\quad l\log p<\log s.$$

Therefore,

$$\beta_p(s)\sim\log 2\cdot\frac{s}{J}\sum_{j=1}^{J}(M-2\delta_j)\quad\text{as } s\to\infty.$$ □

11 p-Adic estimates

In this section, we will determine the p-adic norm of T_n (defined by (8.13)).

By Equation (8.12), Lemma 7.4, and Lemma 8.1(1)(2), we have

$$T_{n,\nu/p^l}=-\mathcal{L}_1\left(V_n\left(t+\frac{\nu}{p^l}\right)\right)-W_n\left(\frac{\nu}{p^l}\right)\tag{11.1}$$

for any integer $\nu\in[1,p^l]$ with $p\nmid\nu$. Substituting (11.1) into (8.13), we have

$$T_n=-\sum_{\substack{1\leqslant\nu\leqslant p^l\\ p\nmid\nu}}\mathcal{L}_1\left(V_n\left(t+\frac{\nu}{p^l}\right)\right)-\sum_{\substack{1\leqslant\nu\leqslant p^l\\ p\nmid\nu}}W_n\left(\frac{\nu}{p^l}\right).\tag{11.2}$$

In the rest of this section, we will show that

- For any sufficiently large $n\in\mathbb{Z}_{>0}$, we can explicitly determine the p-adic order of $\sum_\nu W_n(\nu/p^l)$.
- For any sufficiently large $n\in\mathbb{Z}_{>0}$, the p-adic order of $\mathcal{L}_1\left(V_n\left(t+\nu/p^l\right)\right)$ is strictly larger than the p-adic order of $\sum_\nu W_n(\nu/p^l)$.

In this way, we can explicitly determine $|T_n|_p$ for any sufficiently large $n\in\mathbb{Z}_{>0}$.

We need an elementary lemma:

Lemma 11.1. *For any $m\in\mathbb{Z}$, we have*

$$v_p\left(\sum_{\substack{1\leqslant\nu\leqslant p^l\\ p\nmid\nu}}\nu^m\right)\geqslant l-1.$$

Proof. For any positive integer N, for any integer $\nu \in [1, p^l]$ with $p \nmid \nu$, we have

$$\nu^{N\varphi(p^l)+m} \equiv \nu^m \pmod{p^l\mathbb{Z}_p}.$$

Replacing m by $N\varphi(p^l)+m$ for some sufficiently large positive integer N, we may assume without loss of generality that $m > 0$. Then, the conclusion follows by an easy induction on l. □

Now, we explicitly determine the p-adic order of $\sum_\nu W_n(\nu/p^l)$.

Lemma 11.2. *For any sufficiently large $n \in \mathbb{Z}_{>0}$, we have*

$$\begin{aligned}
&v_p\left(\sum_{\substack{1\leqslant\nu\leqslant p^l\\ p\nmid\nu}} W_n\left(\frac{\nu}{p^l}\right)\right)\\
&= \left(l+\frac{1}{p-1}\right)M\varphi(p^l)n + \frac{s}{J}\sum_{j=1}^{J} v_p(((M-2\delta_j)n)!)\\
&\quad - M\varphi(p^l)v_p(n!) - \left\lfloor\frac{w_n}{n}\right\rfloor v_p(n!) - v_p\left(\left(w_n - \left\lfloor\frac{w_n}{n}\right\rfloor n\right)!\right)\\
&\quad + l|\deg V_n| - v_p(|\deg V_n|-1) - 1. \qquad (11.3)
\end{aligned}$$

(*The main term on the right-hand side is $l|\deg V_n|$*). *In particular, we have*

$$\exp\left(-\alpha_{p,1}(s)n + o(n)\right) \leqslant \left|\sum_{\substack{1\leqslant\nu\leqslant p^l\\ p\nmid\nu}} W_n\left(\frac{\nu}{p^l}\right)\right|_p \leqslant \exp\left(-\alpha_{p,2}(s)n + o(n)\right) \qquad (11.4)$$

as $n \to \infty$, where

$$\alpha_{p,1}(s) = \frac{sl\log p}{J}\sum_{j=1}^{J}(M-2\delta_j) + \frac{s\log p}{(p-1)J}\sum_{j=1}^{J}(M-2\delta_j), \qquad (11.5)$$

$$\alpha_{p,2}(s) = \alpha_{p,1}(s) - \left(l+\frac{1}{p-1}\right)p^{l+3}M\log p. \qquad (11.6)$$

Proof. By (8.6), we have

$$\begin{aligned}
&V_n\left(\frac{t}{p^l}\right)\\
&= p^{(l+1/(p-1))M\varphi(p^l)n}\cdot\frac{\prod_{j=1}^{J}((M-2\delta_j)n)!^{s/J}}{n!^{M\varphi(p^l)}}\cdot\frac{1}{n!^{\lfloor w_n/n\rfloor}(w_n-\lfloor w_n/n\rfloor n)!}\\
&\quad\times p^{l|\deg V_n|}\cdot t^{-|\deg V_n|}\\
&\quad\times\prod_{\substack{1\leqslant\nu\leqslant p^l\\ p\nmid\nu}}\prod_{\tau=0}^{Mn-1}\left(1+(\nu+p^l\tau)t^{-1}\right) \qquad (11.7)
\end{aligned}$$

$$\times \frac{\prod_{\tau=1}^{w_n}\left(1-\tau p^l t^{-1}\right)}{\prod_{j=1}^{J}\prod_{\tau=0}^{(M-2\delta_j)n}\left(1+(\delta_j n+\tau)p^l t^{-1}\right)^{s/J}}. \tag{11.8}$$

Note that the product on line (11.7) is a $\mathbb{Z}_p$-coefficient polynomial in t^{-1} of degree $M\varphi(p^l)n$. The product on line (11.8) belongs to $\mathbb{Z}_p[\![p^l t^{-1}]\!]$. Therefore, we can write $V_n(t/p^l)$ as

$$\begin{aligned} &V_n\left(\frac{t}{p^l}\right)\\ &= p^{(l+1/(p-1))M\varphi(p^l)n}\cdot\frac{\prod_{j=1}^{J}((M-2\delta_j)n)!^{s/J}}{n!^{M\varphi(p^l)}}\cdot\frac{1}{n!^{\lfloor w_n/n\rfloor}(w_n-\lfloor w_n/n\rfloor n)!}\\ &\quad\times p^{l|\deg V_n|}\cdot t^{-|\deg V_n|}\sum_{k=0}^{\infty}h_k t^{-k}, \end{aligned} \tag{11.9}$$

where the coefficients $h_k\in\mathbb{Z}_p$ $(k=0,1,2,\ldots)$ satisfy

$$v_p(h_k)\geqslant l\max\left\{0,k-M\varphi(p^l)n\right\}. \tag{11.10}$$

By Lemma 8.1(3), the function $W_n(t/p^l)$ can be expressed as a $\mathbb{Q}_p$-coefficient convergent power series in t^{-1} for $|t|_p\geqslant 1$ with vanishing constant term. Moreover, by Lemma 8.1(1), we have

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(W_n\left(\frac{t}{p^l}\right)\right)=\frac{1}{p^l}V_n\left(\frac{t}{p^l}\right),\quad |t|_p\geqslant 1. \tag{11.11}$$

Comparing (11.11) to (11.9), we obtain

$$\begin{aligned} &\sum_{\substack{1\leqslant\nu\leqslant p^l\\ p\nmid\nu}}W_n\left(\frac{\nu}{p^l}\right)\\ &= p^{(l+1/(p-1))M\varphi(p^l)n}\cdot\frac{\prod_{j=1}^{J}((M-2\delta_j)n)!^{s/J}}{n!^{M\varphi(p^l)}}\cdot\frac{1}{n!^{\lfloor w_n/n\rfloor}(w_n-\lfloor w_n/n\rfloor n)!}\\ &\quad\times p^{l|\deg V_n|}\cdot\frac{1}{p^l}\\ &\quad\times\sum_{k=0}^{\infty}\frac{h_k}{-|\deg V_n|-k+1}\sum_{\substack{1\leqslant\nu\leqslant p^l\\ p\nmid\nu}}\nu^{-|\deg V_n|-k+1}. \end{aligned} \tag{11.12}$$

In the following, we will prove that for any sufficiently large $n\in\mathbb{Z}_{>0}$

$$\begin{aligned} &v_p\left(\frac{h_k}{-|\deg V_n|-k+1}\sum_{\substack{1\leqslant\nu\leqslant p^l\\ p\nmid\nu}}\nu^{-|\deg V_n|-k+1}\right)\\ &>v_p\left(\frac{h_0}{-|\deg V_n|+1}\sum_{\substack{1\leqslant\nu\leqslant p^l\\ p\nmid\nu}}\nu^{-|\deg V_n|+1}\right)\quad\text{for any } k>0. \end{aligned} \tag{11.13}$$

Note that $h_0 = 1$. By (8.7) and (8.4), we have

$$-|\deg V_n| + 1 \equiv 0 \pmod{(p-1)p^{L_n}}. \tag{11.14}$$

Since $L_n > l$ by (8.3), we have $\varphi(p^l) \mid (-|\deg V_n| + 1)$. Thus,

$$\sum_{\substack{1 \leqslant \nu \leqslant p^l \\ p \nmid \nu}} \nu^{-|\deg V_n|+1} \equiv \sum_{\substack{1 \leqslant \nu \leqslant p^l \\ p \nmid \nu}} 1 \equiv p^{l-1}(p-1) \pmod{p^l \mathbb{Z}_p}.$$

Therefore,

$$v_p\left(\frac{h_0}{-|\deg V_n|+1} \sum_{\substack{1 \leqslant \nu \leqslant p^l \\ p \nmid \nu}} \nu^{-|\deg V_n|+1}\right) = l - 1 - v_p\left(|\deg V_n| - 1\right). \tag{11.15}$$

For $k > 0$, we consider two cases.

Case 1: $v_p(k) < v_p(|\deg V_n| - 1)$. In this case, we have $v_p(-|\deg V_n| - k + 1) = v_p(k) < v_p(|\deg V_n| - 1)$. By (11.10) and Lemma 11.1, we have $v_p(h_k) \geqslant 0$ and

$$v_p\left(\sum_{\substack{1 \leqslant \nu \leqslant p^l \\ p \nmid \nu}} \nu^{-|\deg V_n|-k+1}\right) \geqslant l - 1.$$

In conclusion, (11.13) holds in Case 1.

Case 2: $v_p(k) \geqslant v_p(|\deg V_n| - 1)$. In this case, by (11.14) we have $v_p(k) \geqslant L_n$. By $k > 0$ and (8.3), we have

$$k \geqslant p^{L_n} > p \cdot M\varphi(p^l)n \geqslant 2 \cdot M\varphi(p^l)n. \tag{11.16}$$

By (11.10), we have

$$v_p(h_k) \geqslant l \cdot (k - M\varphi(p^l)n) > 2 \cdot \frac{k}{2} = k. \tag{11.17}$$

Combining (11.17) and Lemma 11.1, we have

$$\begin{aligned} &v_p\left(\frac{h_k}{-|\deg V_n| - k + 1} \sum_{\substack{1 \leqslant \nu \leqslant p^l \\ p \nmid \nu}} \nu^{-|\deg V_n|-k+1}\right) \\ &> k - v_p(|\deg V_n| + k - 1) + l - 1. \end{aligned}$$

In view of (11.15), to prove (11.13), it suffices to prove that

$$k + v_p\left(|\deg V_n| - 1\right) > v_p(|\deg V_n| + k - 1).$$

Since $v_p\left(|\deg V_n| - 1\right) \geqslant L_n > 0$, it suffices to prove that

$$k > \frac{\log(|\deg V_n| + k - 1)}{\log p},$$

or

$$p^k > |\deg V_n| + k - 1.$$

By (8.7), we have $|\deg V_n| + k - 1 < Msn + k$ for sufficiently large n. Note that $k > n$ by (11.16). If n is large enough, we have

$$2^k > Msn + k \quad \text{for any } k > n.$$

Therefore, the claimed inequality (11.13) holds for any sufficiently large n.

By (11.12), (11.13), and (11.15), we obtain an explicit formula for $v_p(\sum_\nu W_n(\nu/p^l))$ for any sufficiently large $n \in \mathbb{Z}_{>0}$, as stated in (11.3). Since $v_p(n!) = n/(p-1) + O(\log n)$ and $0 \leqslant w_n < p^{l+3}Mn$ (see (8.5)), Equation (11.3) implies Inequality (11.4). The proof of Lemma 11.2 is complete. □

Next, we bound from below the p-adic order of $\mathcal{L}_1(V_n(t + \nu/p^l))$.

Lemma 11.3. *For any $n \in \mathbb{Z}_{>0}$, for any integer $\nu \in [1, p^l]$ with $p \nmid \nu$, we have*

$$\begin{aligned}
&v_p\left(\mathcal{L}_1\left(V_n\left(t + \frac{\nu}{p^l}\right)\right)\right) \\
\geqslant\ &\left(l + \frac{1}{p-1}\right) M\varphi(p^l)n + \frac{s}{J}\sum_{j=1}^{J} v_p(((M - 2\delta_j)n)!) \\
&- M\varphi(p^l)v_p(n!) - \left\lfloor \frac{w_n}{n} \right\rfloor v_p(n!) - v_p\left(\left(w_n - \left\lfloor \frac{w_n}{n} \right\rfloor n\right)!\right) \\
&+ l|\deg V_n| + lMn - 1 - \frac{\log(Mn+1)}{\log p}.
\end{aligned} \tag{11.18}$$

Proof. By (8.6), we have

$$\begin{aligned}
&V_n\left(t + \frac{\nu}{p^l}\right) \\
&= p^{(l+1/(p-1))M\varphi(p^l)n} \cdot \frac{\prod_{j=1}^{J}((M - 2\delta_j)n)!^{s/J}}{n!^{M\varphi(p^l)}} \cdot \frac{1}{n!^{\lfloor w_n/n \rfloor}(w_n - \lfloor w_n/n \rfloor n)!} \\
&\quad \times p^{l|\deg V_n|} \cdot p^{lMn} \\
&\quad \times \prod_{\tau=0}^{Mn-1}(t + \tau + 1) \cdot \prod_{\substack{1 \leqslant \nu' \leqslant p^l \\ p \nmid \nu',\ \nu' \neq p^l - \nu}} \prod_{\tau=0}^{Mn-1}(p^l t + p^l\tau + \nu + \nu') \qquad (11.19) \\
&\quad \times \frac{\prod_{\tau=1}^{w_n}(p^l t - p^l\tau + \nu)}{\prod_{j=1}^{J}\prod_{\tau=0}^{(M-2\delta_j)n}(p^l t + p^l(\delta_j n + \tau) + \nu)^{s/J}}. \qquad (11.20)
\end{aligned}$$

The first product on line (11.19) is a $\mathbb{Z}_p$-coefficient polynomial in t of degree Mn. The second product on line (11.19) and the product on line (11.20) both belong to $\mathbb{Z}_p[\![p^l t]\!]$. Therefore, we can write $V_n(t + \nu/p^l)$ as

$$V_n\left(t + \frac{\nu}{p^l}\right)$$

$$= p^{(l+1/(p-1))M\varphi(p^l)n} \cdot \frac{\prod_{j=1}^{J}((M-2\delta_j)n)!^{s/J}}{n!^{M\varphi(p^l)}} \cdot \frac{1}{n!^{\lfloor w_n/n \rfloor}(w_n - \lfloor w_n/n \rfloor n)!}$$
$$\times p^{l|\deg V_n|} \cdot p^{lMn} \cdot \sum_{k=0}^{\infty} u_k t^k, \tag{11.21}$$

where the coefficients $u_k \in \mathbb{Z}_p$ $(k = 0, 1, 2, \ldots)$ satisfy

$$v_p(u_k) \geqslant l \max\{0, k - Mn\}. \tag{11.22}$$

By (11.21) and Definition 7.3, we have

$$\mathcal{L}_1\left(V_n\left(t + \frac{\nu}{p^l}\right)\right)$$
$$= p^{(l+1/(p-1))M\varphi(p^l)n} \cdot \frac{\prod_{j=1}^{J}((M-2\delta_j)n)!^{s/J}}{n!^{M\varphi(p^l)}} \cdot \frac{1}{n!^{\lfloor w_n/n \rfloor}(w_n - \lfloor w_n/n \rfloor n)!}$$
$$\times p^{l|\deg V_n|} \cdot p^{lMn} \cdot \sum_{k=0}^{\infty} \frac{u_k B_{k+1}}{k+1}. \tag{11.23}$$

By (11.22) and the von Staudt–Clausen theorem, we have

$$\begin{aligned} &\inf_{k \geqslant 0} v_p\left(\frac{u_k B_{k+1}}{k+1}\right) \\ \geqslant\ &\inf_{k \geqslant 0}\left(-1 - \frac{\log(k+1)}{\log p} + \max\{0, k - Mn\}\right) \\ =\ &-1 - \frac{\log(Mn+1)}{\log p}. \end{aligned} \tag{11.24}$$

By (11.23) and (11.24), we obtain (11.18). The proof of Lemma 11.3 is complete. □

Now, we can determine $|T_n|_p$ explicitly for any sufficiently large $n \in \mathbb{Z}_{>0}$.

Lemma 11.4. *We have*

$$\exp\left(-\alpha_{p,1}(s)n + o(n)\right) \leqslant |T_n|_p \leqslant \exp\left(-\alpha_{p,2}(s)n + o(n)\right) \quad \text{as } n \to \infty, \tag{11.25}$$

where $\alpha_{p,1}(s), \alpha_{p,2}(s)$ are given by (11.5) *and* (11.6)*, respectively. Moreover, we have*

$$\alpha_{p,1}(s) \sim \frac{s \log s}{J} \sum_{j=1}^{J}(M - 2\delta_j) \quad \text{as } s \to \infty, \tag{11.26}$$

$$\alpha_{p,1}(s) - \alpha_{p,2}(s) = O\left(\frac{s}{\log s}\right) \quad \text{as } s \to \infty. \tag{11.27}$$

Proof. By Lemmas 11.2 and 11.3, and noting that

$$lMn - 1 - \frac{\log(Mn+1)}{\log p} > -v_p(|\deg V_n| - 1) - 1$$

for any sufficiently large $n \in \mathbb{Z}_{>0}$, we obtain

$$v_p\left(\sum_{\substack{1 \leqslant \nu \leqslant p^l \\ p \nmid \nu}} \mathcal{L}_1\left(V_n\left(t + \frac{\nu}{p^l}\right)\right)\right) > v_p\left(\sum_{\substack{1 \leqslant \nu \leqslant p^l \\ p \nmid \nu}} W_n\left(\frac{\nu}{p^l}\right)\right). \tag{11.28}$$

Combining (11.28) and (11.2), we have

$$|T_n|_p = \left|\sum_{\substack{1 \leqslant \nu \leqslant p^l \\ p \nmid \nu}} W_n\left(\frac{\nu}{p^l}\right)\right|_p$$

for any sufficiently large $n \in \mathbb{Z}_{>0}$. Therefore, (11.25) follows from (11.4).

By (8.2), we have

$$l \log p = \log s + O(\log\log s) \quad \text{as } s \to \infty,$$
$$lp^l = O\left(\frac{s}{\log s}\right) \quad \text{as } s \to \infty.$$

Therefore, the asymptotic estimates for $\alpha_{p,1}(s)$ and $\alpha_{p,1}(s) - \alpha_{p,2}(s)$ follow. □

12 Proof of Theorem 1.4

We first reduce the proof of Theorem 1.4 to a computational problem.

Proposition 12.1. *Let $\psi(\cdot)$ be the digamma function. For any collection of nonnegative integers $(M, \delta_1, \delta_2, \ldots, \delta_J)$ under the constraints $J \geqslant 1$ and*

$$0 \leqslant \delta_1 \leqslant \delta_2 \leqslant \cdots \leqslant \delta_J < \frac{M}{2},$$

we define

$$\varpi = \int_0^1 \phi(x)\ \mathrm{d}\psi(x) - \int_0^{1/(M-2\delta_1)} \phi(x)\ \frac{\mathrm{d}x}{x^2},$$

where the function $\phi(\cdot)$ is given by

$$\phi(x) = \inf_{y \in \mathbb{R}} \sum_{j=1}^{J} \left(\lfloor (M-2\delta_j)x \rfloor - \lfloor y - \delta_j x \rfloor - \lfloor (M-\delta_j)x - y \rfloor\right).$$

Then, we have

$$\dim_{\mathbb{Q}} \mathrm{Span}_{\mathbb{Q}}\left(1, \zeta_p(3), \zeta_p(5), \ldots, \zeta_p(s-1)\right) \geqslant (C' - o(1)) \cdot \log s$$

as the positive even integer $s \to \infty$, where the constant C' is given by

$$C' = \frac{\sum_{j=1}^{J}(M-2\delta_j)}{\log 2 \cdot \sum_{j=1}^{J}(M-2\delta_j) - \varpi + J\cdot(M-2\delta_1)}. \tag{12.1}$$

Proof. For any integer $n > s^2$, consider

$$\widehat{T}_n := \Phi_n^{-s/J} D^s_{(M-2\delta_1)n} \cdot T_n.$$

By Lemmas 8.3 and 9.2,

$$\widehat{T}_n = \widehat{\sigma}_{n,0} + \sum_{\substack{3\leqslant i\leqslant s-1 \\ i \text{ odd}}} \widehat{\sigma}_{n,i}\zeta_p(i)$$

is a linear form in 1 and p-adic odd zeta values with integer coefficients

$$\widehat{\sigma}_{n,i} := \Phi_n^{-s/J} D^s_{(M-2\delta_1)n} \cdot \sigma_{n,i} \cdot p^{li} \in \mathbb{Z}, \quad i = 0,3,5,\ldots,s-1.$$

By Lemma 11.4 and the facts $v_p(\Phi_n) = 0$ for $n > p^2$ and $v_p(D_{(M-2\delta_1)n}) = O(\log n)$, we have

$$\exp\left(-\widehat{\alpha}_{p,1}(s)n + o(n)\right) \leqslant |\widehat{T}_n|_p \leqslant \exp\left(-\widehat{\alpha}_{p,2}(s)n + o(n)\right) \quad \text{as } n \to \infty,$$

where

$$\widehat{\alpha}_{p,j}(s) = \alpha_{p,j}(s), \quad j = 1,2.$$

By Lemma 10.1, Equation (5.18), and Lemma 5.3, we have

$$\max_{i=0,3,5,\ldots,s-1} |\widehat{\sigma}_{n,i}| \leqslant \exp\left(\widehat{\beta}_p(s)n + o(n)\right) \quad \text{as } n \to \infty,$$

where

$$\widehat{\beta}_p(s) = \beta_p(s) - \frac{s}{J}\varpi + s(M-2\delta_1).$$

By (11.26), (11.27), and (10.2), we have the following asymptotic estimates for $\widehat{\alpha}_{p,j}(s)$ and $\widehat{\beta}_p(s)$ as $s \to \infty$:

$$\widehat{\alpha}_{p,1}(s) \sim \frac{s\log s}{J}\sum_{j=1}^{J}(M-2\delta_j), \tag{12.2}$$

$$\widehat{\alpha}_{p,1}(s) - \widehat{\alpha}_{p,2}(s) = O\left(\frac{s}{\log s}\right), \tag{12.3}$$

$$\widehat{\beta}_p(s) \sim \log 2 \cdot \frac{s}{J}\sum_{j=1}^{J}(M-2\delta_j) - \frac{s}{J}\varpi + s(M-2\delta_1). \tag{12.4}$$

Applying Nesterenko's p-adic linear independence criterion (Theorem 2.3) to the sequence of linear forms $\{\widehat{T}_n\}_{n>s^2}$, and using (12.2)–(12.4), we obtain

$$\dim_{\mathbb{Q}} \operatorname{Span}_{\mathbb{Q}}\left(1, \zeta_p(3), \zeta_p(5), \ldots, \zeta_p(s-1)\right) \geqslant \frac{\widehat{\alpha}_{p,1}(s)}{\widehat{\beta}_p(s) + \widehat{\alpha}_{p,1}(s) - \widehat{\alpha}_{p,2}(s)}$$

$$= (C' - o(1)) \cdot \log s \quad \text{as } s \to \infty, \tag{12.5}$$

where the constant C' is given by

$$C' = \frac{\sum_{j=1}^{J}(M - 2\delta_j)}{\log 2 \cdot \sum_{j=1}^{J}(M - 2\delta_j) - \varpi + J \cdot (M - 2\delta_1)}.$$

We have restricted s to $2J\mathbb{Z}_{>0}$ in our constructions of rational functions and linear forms. Thus, (12.5) holds as $s \in 2J\mathbb{Z}_{>0}$ and $s \to \infty$. However, if we replace the condition '$s \in 2J\mathbb{Z}_{>0}$' by '$s \in 2\mathbb{Z}_{>0}$', inequality (12.5) remains valid. □

If we take $M = 1$, $\delta_1 = 0$ (with $J = 1$) in Proposition 12.1, then we find $\phi(x) \equiv 0$, $\varpi = 0$, and $C' = 1/(1 + \log 2)$. Thus, we rediscover Sprang's theorem [16]. A simple choice of parameters to improve Sprang's theorem is: $M = 6$, $\delta_1 = 0$, $\delta_2 = 1$ (with $J = 2$). This collection of parameters leads to the proof of Theorem 1.4.

Proof of Theorem 1.4. Take $M = 6$, $\delta_1 = 0$, $\delta_2 = 1$ (with $J = 2$) in Proposition 12.1. As calculated in the proof of Theorem 1.3 (see Section 6), $\varpi = 2.157479\ldots$. Now, by (12.1), we have

$$C' = \frac{10}{10 \log 2 - \varpi + 12} = \frac{1.009388\ldots}{1 + \log 2}.$$

In conclusion, by Proposition 12.1, we have

$$\dim_{\mathbb{Q}} \mathrm{Span}_{\mathbb{Q}} \left(1, \zeta_p(3), \zeta_p(5), \ldots, \zeta_p(s-1)\right) \geqslant \frac{1.009}{1 + \log 2} \cdot \log s$$

for any sufficiently large even integer s. The proof of Theorem 1.4 is complete. □

13 Computational explorations

In this section, we will present computational results and prove Claims 1.5 and 1.6.

Recall that Proposition 6.1 and Proposition 12.1 reduce the theoretic problems to finite computations. Since the dependence of ϖ (and hence C, C') on the parameters is discrete, it is unlikely to find an optimal constant C for Proposition 6.1 or an optimal constant C' for Proposition 12.1.

The parameters $M = 6$, $\delta_1 = 0$, $\delta_2 = 1$ (with $J = 2$) that we used in the proofs of Theorems 1.3 and 1.4 are very simple. Other simple parameter choices improve the constant $1.009/(1 + \log 2)$ in Theorems 1.3 and 1.4:

- $M = 19$, $(\delta_1, \delta_2, \delta_3) = (0, 1, 2)$. Then, $\varpi = 9.023331\ldots$,

$$C = C' = \frac{1.036282\ldots}{1 + \log 2}.$$

- $M = 12$, $(\delta_1, \ldots, \delta_4) = (0, 0, 1, 2)$. Then, $\varpi = 9.363813\ldots$,

$$C = C' = \frac{1.049651\ldots}{1 + \log 2}.$$

- $M = 16$, $(\delta_1, \dots, \delta_5) = (0, 0, 1, 2, 3)$. Then, $\varpi = 18.818264\dots$,

$$C = C' = \frac{1.062948\dots}{1 + \log 2}.$$

- $M = 37$, $(\delta_1, \dots, \delta_{10}) = (2, 3, \dots, 11)$. Then, $\varpi = 103.306060\dots$,

$$C = \frac{1.026022\dots}{1 + \log 2}, \quad C' = \frac{1.033853\dots}{1 + \log 2}.$$

  (The last set of parameters is of historical significance. Zudilin used them in [17] to prove that at least one of $\zeta(5), \zeta(7), \zeta(9), \zeta(11)$ is irrational.)

Remark 13.1. If $\delta_1 = 0$, then by (6.1) and (12.1) we have $C = C'$.

If we replace $(M; \delta_1, \dots, \delta_J)$ by

$$(\widetilde{M}; \widetilde{\delta}_1, \widetilde{\delta}_2, \dots, \widetilde{\delta}_J) = (M - 2\delta_1; \delta_1 - \delta_1, \delta_2 - \delta_1, \dots, \delta_J - \delta_1),$$

then $\widetilde{\varpi} = \varpi$, and $\widetilde{C} = \widetilde{C'} = C' \geqslant C$. We may therefore assume $\delta_1 = 0$ without decreasing C and C'.

After an extensive random search in the range $M \leqslant 600$, $\delta_j \leqslant 200$ $(j = 1, 2, \dots, J)$, and $J \leqslant 100$, we have found a better constant $C = C'$ to be approximately $1.119/(1 + \log 2)$. For an explanation of how to calculate ϖ using a `MATLAB` program, we refer the reader to [10, §5]. Now, we give the proof of Claim 1.5.

Proof of Claim 1.5. Take the parameters $M = 433$, $J = 89$, and $\delta_1, \dots, \delta_{89}$ in Table 1. Then, we have $\varpi = 12557.653439\dots$ and

$\delta_1 = 0$	$\delta_2 = 0$	$\delta_3 = 0$	$\delta_4 = 0$	$\delta_5 = 0$	$\delta_6 = 0$	$\delta_7 = 0$
$\delta_8 = 0$	$\delta_9 = 0$	$\delta_{10} = 0$	$\delta_{11} = 0$	$\delta_{12} = 0$	$\delta_{13} = 0$	$\delta_{14} = 0$
$\delta_{15} = 0$	$\delta_{16} = 0$	$\delta_{17} = 0$	$\delta_{18} = 0$	$\delta_{19} = 1$	$\delta_{20} = 1$	$\delta_{21} = 2$
$\delta_{22} = 2$	$\delta_{23} = 3$	$\delta_{24} = 3$	$\delta_{25} = 4$	$\delta_{26} = 5$	$\delta_{27} = 6$	$\delta_{28} = 7$
$\delta_{29} = 8$	$\delta_{30} = 9$	$\delta_{31} = 10$	$\delta_{32} = 11$	$\delta_{33} = 12$	$\delta_{34} = 13$	$\delta_{35} = 14$
$\delta_{36} = 15$	$\delta_{37} = 16$	$\delta_{38} = 17$	$\delta_{39} = 18$	$\delta_{40} = 19$	$\delta_{41} = 20$	$\delta_{42} = 21$
$\delta_{43} = 22$	$\delta_{44} = 23$	$\delta_{45} = 25$	$\delta_{46} = 27$	$\delta_{47} = 29$	$\delta_{48} = 31$	$\delta_{49} = 33$
$\delta_{50} = 35$	$\delta_{51} = 37$	$\delta_{52} = 39$	$\delta_{53} = 41$	$\delta_{54} = 43$	$\delta_{55} = 45$	$\delta_{56} = 47$
$\delta_{57} = 49$	$\delta_{58} = 51$	$\delta_{59} = 53$	$\delta_{60} = 55$	$\delta_{61} = 57$	$\delta_{62} = 59$	$\delta_{63} = 61$
$\delta_{64} = 63$	$\delta_{65} = 65$	$\delta_{66} = 67$	$\delta_{67} = 69$	$\delta_{68} = 71$	$\delta_{69} = 73$	$\delta_{70} = 75$
$\delta_{71} = 77$	$\delta_{72} = 79$	$\delta_{73} = 81$	$\delta_{74} = 83$	$\delta_{75} = 85$	$\delta_{76} = 87$	$\delta_{77} = 89$
$\delta_{78} = 91$	$\delta_{79} = 93$	$\delta_{80} = 95$	$\delta_{81} = 97$	$\delta_{82} = 99$	$\delta_{83} = 101$	$\delta_{84} = 103$
$\delta_{85} = 107$	$\delta_{86} = 111$	$\delta_{87} = 115$	$\delta_{88} = 119$	$\delta_{89} = 123$		

Table 1: The choice of $(\delta_1, \dots, \delta_{89})$

$$C = C' = \frac{1.119356\dots}{1 + \log 2}.$$

Claim 1.5 now follows from Propositions 6.1 and 12.1. □

As a byproduct of their work [2], Ball and Rivoal proved that

$$\dim_{\mathbb{Q}} \operatorname{Span}_{\mathbb{Q}} (1, \zeta(3), \zeta(5), \dots, \zeta(169)) \geqslant 3.$$

In other words, we have the upper bound $\kappa_3 \leqslant 169$ for the least odd integer $\kappa_3 \geqslant 5$ such that $\dim_{\mathbb{Q}} \operatorname{Span}_{\mathbb{Q}} (1, \zeta(3), \zeta(5), \dots, \zeta(\kappa_3)) = 3$. Of course, the standard conjecture is $\kappa_3 = 5$. The upper bound for κ_3 was improved by Zudilin [18] to 145 and then improved further to 139 by Fischler and Zudilin [7]. Using more elaborate parameters, we prove that $\kappa_3 \leqslant 75$.

A sketch of the proof of Claim 1.6. We use the following rational functions:

$$\begin{aligned}\widetilde{R}_n(t) =& \frac{\prod_{j=1}^{76}((M-2\delta_j)n)!}{n!^{2r}} \cdot (2t+Mn) \\ & \times \frac{(t-rn)_{rn}(t+Mn+1)_{rn}}{\prod_{j=1}^{76}(t+\delta_j n)_{(M-2\delta_j)n+1}}, \quad n = 1, 2, 3, \dots,\end{aligned}$$

where $r = 2444$, $M = 444$, and $\delta_1, \dots, \delta_{76}$ are those in Table 2.

$\delta_1 = 1$	$\delta_{12} = 5$	$\delta_{23} = 16$	$\delta_{34} = 30$	$\delta_{45} = 52$	$\delta_{56} = 74$	$\delta_{67} = 96$
$\delta_2 = 1$	$\delta_{13} = 6$	$\delta_{24} = 17$	$\delta_{35} = 32$	$\delta_{46} = 54$	$\delta_{57} = 76$	$\delta_{68} = 98$
$\delta_3 = 1$	$\delta_{14} = 7$	$\delta_{25} = 18$	$\delta_{36} = 34$	$\delta_{47} = 56$	$\delta_{58} = 78$	$\delta_{69} = 100$
$\delta_4 = 1$	$\delta_{15} = 8$	$\delta_{26} = 19$	$\delta_{37} = 36$	$\delta_{48} = 58$	$\delta_{59} = 80$	$\delta_{70} = 102$
$\delta_5 = 1$	$\delta_{16} = 9$	$\delta_{27} = 20$	$\delta_{38} = 38$	$\delta_{49} = 60$	$\delta_{60} = 82$	$\delta_{71} = 104$
$\delta_6 = 2$	$\delta_{17} = 10$	$\delta_{28} = 21$	$\delta_{39} = 40$	$\delta_{50} = 62$	$\delta_{61} = 84$	$\delta_{72} = 108$
$\delta_7 = 2$	$\delta_{18} = 11$	$\delta_{29} = 22$	$\delta_{40} = 42$	$\delta_{51} = 64$	$\delta_{62} = 86$	$\delta_{73} = 112$
$\delta_8 = 3$	$\delta_{19} = 12$	$\delta_{30} = 23$	$\delta_{41} = 44$	$\delta_{52} = 66$	$\delta_{63} = 88$	$\delta_{74} = 116$
$\delta_9 = 3$	$\delta_{20} = 13$	$\delta_{31} = 24$	$\delta_{42} = 46$	$\delta_{53} = 68$	$\delta_{64} = 90$	$\delta_{75} = 120$
$\delta_{10} = 4$	$\delta_{21} = 14$	$\delta_{32} = 26$	$\delta_{43} = 48$	$\delta_{54} = 70$	$\delta_{65} = 92$	$\delta_{76} = 124$
$\delta_{11} = 4$	$\delta_{22} = 15$	$\delta_{33} = 28$	$\delta_{44} = 50$	$\delta_{55} = 72$	$\delta_{66} = 94$	

Table 2: The choice of $(\delta_1, \dots, \delta_{76})$

Then,

$$\widetilde{S}_n := \sum_{\nu=1}^{\infty} \widetilde{R}_n(\nu) = \widetilde{\rho}_{n,0} + \sum_{\substack{3 \leqslant i \leqslant 75 \\ i \text{ odd}}} \widetilde{\rho}_{n,i} \zeta(i)$$

is a linear form in 1, $\zeta(3)$, $\zeta(5)$, …, $\zeta(75)$ with rational coefficients. For any $n > 76^2$ we have

$$\widetilde{\Phi}_n^{-1} \prod_{j=1}^{76} D_{\max\{(M-2\delta_1)n,\ (M-\delta_j)n\}} \cdot \widetilde{\rho}_{n,0} \in \mathbb{Z}$$

and

$$\widetilde{\Phi}_n^{-1} D_{(M-2\delta_1)n}^{76-i} \cdot \widetilde{\rho}_{n,i} \in \mathbb{Z}, \quad i = 3, 5, \dots, 75,$$

where

$$\widetilde{\Phi}_n = \prod_{\substack{\sqrt{(2r+M)n} < q \leqslant (M-2\delta_1)n \\ q \text{ prime}}} q^{\widetilde{\phi}(n/q)},$$

with $\widetilde{\phi}(x) = \min_{0 \leqslant y < 1} \widetilde{\phi}(x, y)$ and

$$\begin{aligned}\widetilde{\phi}(x, y) =& \lfloor rx + y \rfloor + \lfloor (r + M)x - y \rfloor - \lfloor y \rfloor - \lfloor Mx - y \rfloor - 2r\lfloor x \rfloor \\ &+ \sum_{j=1}^{76} \left(\lfloor (M - 2\delta_j)x \rfloor - \lfloor y - \delta_j x \rfloor - \lfloor (M - \delta_j)x - y \rfloor \right).\end{aligned}$$

Note that not only 'denominator type elementary bricks' but also 'numerator type elementary bricks' contribute to $\widetilde{\Phi}_n$.

As $n \to \infty$, we have the asymptotic estimates:

$$\begin{aligned}\widetilde{S}_n &= \exp\left(-\widetilde{\alpha} n + o(n)\right), \\ \widetilde{\Phi}_n &= \exp\left(\widetilde{\varpi} n + o(n)\right),\end{aligned}$$

where

$$\begin{aligned}\widetilde{\varpi} &= \int_0^1 \widetilde{\phi}(x)\, \mathrm{d}\psi(x) - \int_0^{1/(M-2\delta_1)} \widetilde{\phi}(x)\, \frac{\mathrm{d}x}{x^2} \\ &= 42945.452053\ldots,\end{aligned}$$

$$\begin{aligned}\widetilde{\alpha} =& -(2r + M)\log(2r + M + x_0) - r\log(r + x_0) \\ &+ (r + M)\log(r + M + x_0) \\ &- \sum_{j=1}^{76} (M - 2\delta_j)\log(M - 2\delta_j) \\ &- \sum_{j=1}^{76} (r + \delta_j)\log(r + \delta_j + x_0) \\ &+ \sum_{j=1}^{76} (r + M - \delta_j)\log(r + M - \delta_j + x_0) \\ =& \ 38489.009014\ldots,\end{aligned}$$

and $x_0 = 0.194387\ldots$ is the unique positive real root of the polynomial

$$\begin{aligned}&(2r + M + X)(r + X)\prod_{j=1}^{76}(r + \delta_j + X) \\ -\ &X(r + M + X)\prod_{j=1}^{76}(r + M - \delta_j + X).\end{aligned}$$

On the other hand, we have the bound

$$\max_{i=0,3,5,\ldots,75} |\widetilde{\rho}_{n,i}| \leqslant \exp\Big(\widetilde{\beta} n + o(n)\Big) \quad \text{as } n \to \infty,$$

where

$$\widetilde{\beta} = (2r+M)\log\frac{2r+M}{2} - M\log\frac{M}{2} + \log 2\cdot\sum_{j=1}^{76}(M-2\delta_j)$$
$$= 58209.043057\ldots.$$

Applying Theorem 2.2 to the sequence of linear forms

$$\left\{\widetilde{\Phi}_n^{-1}\prod_{j=1}^{76} D_{\max\{(M-2\delta_1)n,\ (M-\delta_j)n\}}\cdot\widetilde{S}_n\right\}_{n>76^2}$$

with the divisor $d_{n,i} = D_{443n}^3$ of $\widetilde{\Phi}_n^{-1}\prod_{j=1}^{76} D_{\max\{(M-2\delta_1)n,\ (M-\delta_j)n\}}\cdot\widetilde{\rho}_{n,2i+1}$ for $i = 1, 2, 3, \ldots, 37$, we obtain the following bound, with $\gamma_1 = 443\times 3$:

$$\begin{aligned}&\dim_{\mathbb{Q}}\operatorname{Span}_{\mathbb{Q}}(1,\zeta(3),\zeta(5),\ldots,\zeta(75))\\ \geqslant& 1+\frac{\widetilde{\alpha}-\sum_{j=1}^{76}\max\{M-2\delta_1,\ M-\delta_j\}+\widetilde{\varpi}+\gamma_1}{\widetilde{\beta}+\sum_{j=1}^{76}\max\{M-2\delta_1,\ M-\delta_j\}-\widetilde{\varpi}}\\ =& 2.006260\ldots.\end{aligned}$$

This completes the proof of Claim 1.6. □

Statement on AI use: This work was completed in 2024; no AI tools were used. During the 2026 revision process, I used ChatGPT (GPT-6 Astra, OpenAI) to check the proof.

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